

Streaming, music & entertainment

AHMEDABAD, IPL FINAL, 7:28 PM

Two minutes before the first ball, crores of phones open JioHotstar at once. Priya, 24, sitting on the steps outside her hostel, gets a spinning wheel for eight seconds, then a blurry picture that sharpens as the first over begins. Her father in Rajkot is watching the same match on television, about 30 seconds ahead of her. He phones to shout about a wicket she hasn't seen yet. The next morning, lakhs of people who paid for a plan in March to watch the IPL start deciding whether to keep paying now that the cricket is over.

Try it: if you ran JioHotstar, which would worry you more, the eight seconds of buffering on final night or the people who cancel in June? Why? Think for five minutes before reading on.

Paying once for a show, earning from it many times

A streaming service pays a huge, fixed sum for its shows, matches and songs, and then tries to earn that money back from as many people as possible: through monthly plans, ads, bundles with a mobile recharge, or all three. One more viewer costs it almost nothing in content, so size wins. In India the audience is enormous but pays little, and the biggest draw, cricket, lasts two months a year. Almost every PM question in this sector comes back to three things: what to pay for content, how to get people to pay or watch ads, and how to keep them once the thing they came for is over.

53 crore

people used JioHotstar each month, April–June 2026, its highest ever (JioStar)

6.5 crore

watched one match on JioHotstar at the same moment: India–England, T20 World Cup semi-final, March 2026 (ICC)

₹149

Netflix's cheapest plan in India, phone only; its 4K plan costs ₹649 a month (2026)

30 crore

people paid for Spotify Premium worldwide, out of 77.7 crore monthly listeners (Q2 2026)

Who does what, and who pays whom

Six groups share the money (Figure 6.1). **Rights owners**, such as the BCCI for the IPL, film studios and record labels, own what people want to watch or hear, and sell the right to show it. The **streaming app** buys those rights, delivers the video and sells plans and ads. **Viewers** pay a monthly fee, or pay nothing and watch ads. **Advertisers** pay for every thousand people who see their ad. **Telecom operators** such as Jio and Airtel pay the app to include it in a recharge, because “free JioHotstar with this pack” sells bigger recharges. And **delivery networks** carry the video to your phone, charging for every gigabyte.

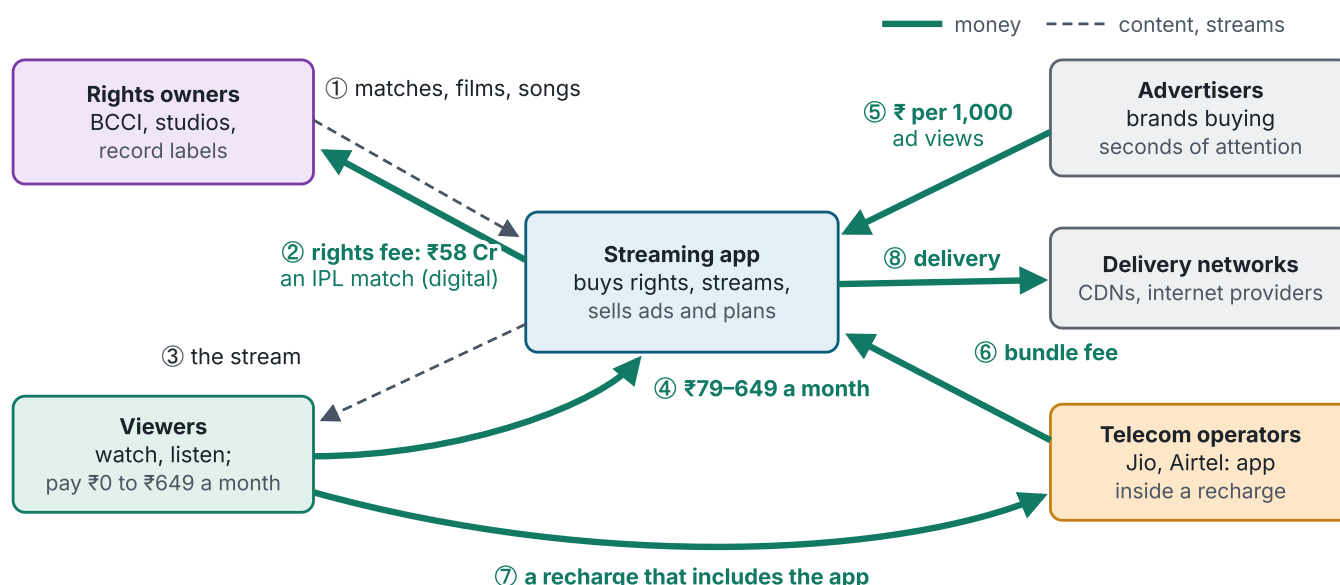


Figure 6.1 • Who pays whom when you watch. The app sits between people who own content and people who pay for it, in money or in attention. The rights fee (②) is fixed; everything on the right has to add up to more than it. *Author's diagram; IPL digital rights from the BCCI e-auction (June 2022); plan prices from Netflix and JioHotstar (2026).*

There are five ways to charge. **SVOD** (subscription video on demand) means a monthly plan with no ads: Netflix's original model. **AVOD** (advertising video on demand) is free to watch and paid for by ads: YouTube. **Hybrid** is a

cheaper plan with ads, now the default at Prime Video India. **TVOD** (transactional video on demand) is paying per film, like renting a DVD used to be. A **bundle** means someone else pays: your Jio recharge, or Amazon Prime. Games add a sixth: free to play, with a few players buying extras.

The unit economics, worked in rupees

Take one person on a ₹199-a-month plan for a year: ₹2,388 (Figure 6.2). The government takes 18% GST, about ₹364. Payment fees take about ₹40. Then comes the share of content cost: shows and films are paid for in advance and spread over the years they are watched, and at Netflix this is the biggest single cost, roughly 40% of revenue. After streaming, marketing and staff, about ₹460 is left (illustrative).

₹199 a month $\times 12 = ₹2,388$. Where it goes (illustrative shares, modelled on Netflix's global cost mix).

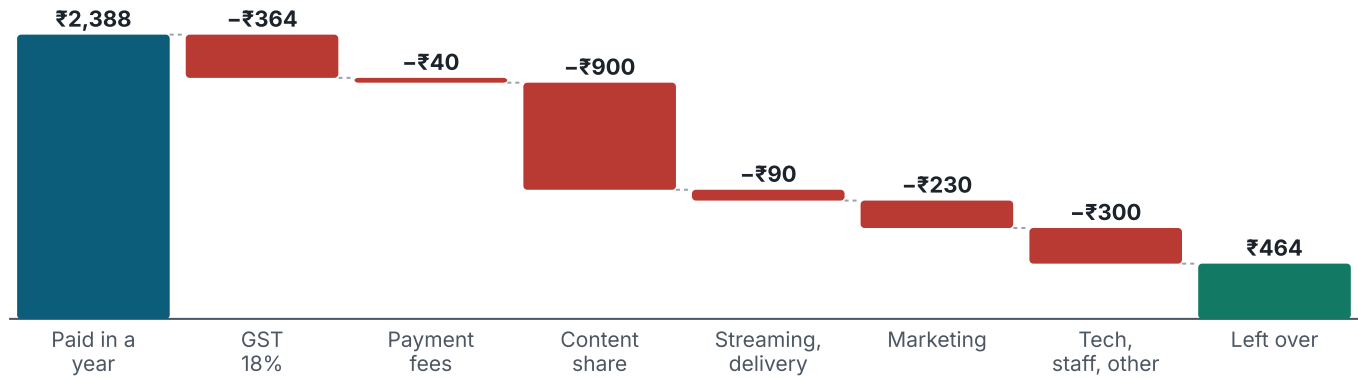


Figure 6.2 • One subscriber's year. Content is the biggest cost, but it is not really "per subscriber": the show costs the same whether 10 lakh or 1 crore people watch it. The shares here assume the service is already large. *Illustrative; cost shares modelled on Netflix's reported global cost structure (2025–26).*

That last point is the key to the whole sector. Content is a **fixed cost** (the same total whatever the number of customers). Suppose a service spends ₹2,000 crore a year on content. With 50 lakh subscribers, each one "costs" $₹2,000 \text{ Cr} \div 50 \text{ lakh} = ₹4,000$ a year in content, far more than they pay. With 4 crore subscribers, the same content costs ₹500 each (Figure 6.3). The business only works past a certain size, which is why streaming companies merge, bundle and spend heavily to grow.

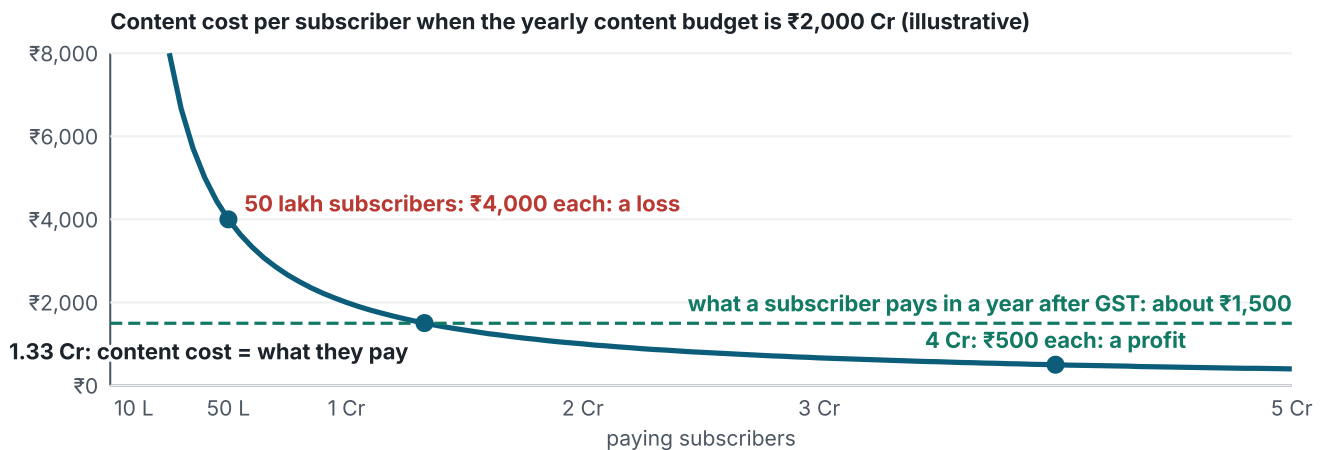
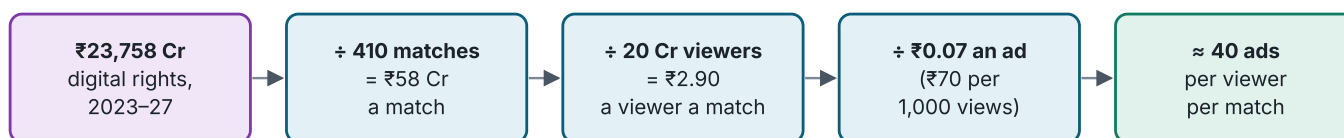


Figure 6.3 • Why scale wins. The content bill is fixed, so the cost per subscriber falls as the audience grows. Below about 1.3 crore paying subscribers this service loses money on content alone; above it, every new subscriber is mostly profit. *Illustrative budget and price.*

Cricket: the most expensive show in India

In June 2022 the BCCI sold IPL rights for 2023–27 in separate packages for the first time. Viacom18 (now part of JioStar) paid **₹23,758 crore for the digital rights** to 410 matches; Star (also now JioStar) paid ₹23,575 crore for television. That is $₹23,758 \text{ Cr} \div 410 = \text{about } ₹58 \text{ crore a match}$, for phones alone. If 20 crore people watch a match, each viewer must bring in ₹2.90 just to cover the rights. At ₹70 per 1,000 ad views, one ad earns 7 paise, so each viewer must see about 40 ads per match (Figure 6.4).

What one IPL match must earn on phones, before a rupee of profit (viewers and ad price illustrative)



40 ads of 10–20 seconds is 7–13 minutes of ads per viewer. That is why every over-break, strategic time-out and wicket is sold, and why the rights price sets the product: miss the ad target and the app must charge viewers instead.

Figure 6.4 · The rights price, turned into ads per viewer. A fixed rights fee becomes a target for every over-break. If ads fall short, the gap has to come from subscriptions or bundles. *Rights value from the BCCI e-auction (June 2022); viewers per match and ad price illustrative.*

Three numbers about viewers are often confused (Figure 6.5). **Reach** is how many different people watched at some point: 70 crore for IPL 2026 on JioHotstar. **Concurrency** is how many watch at the same moment: the record is 6.5 crore. Paying subscriptions are a third number again, and many of them come bundled with a recharge. Ads are sold on reach and views; servers are built for concurrency; revenue per user comes from subscriptions. Headlines that say “82 crore viewers” are usually reach, not people watching at once.

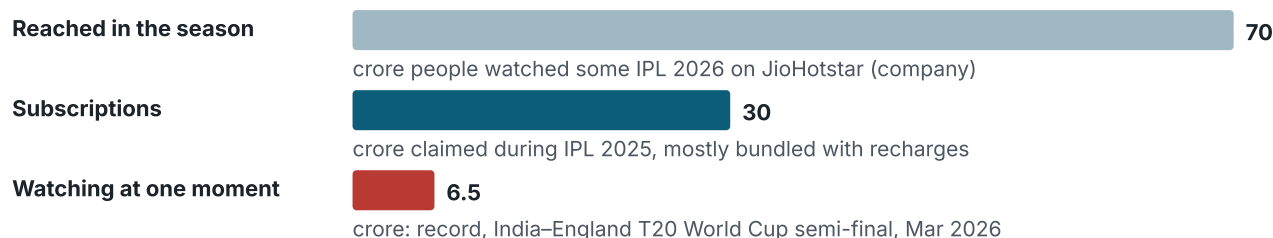


Figure 6.5 · Reach, subscriptions and concurrency are different numbers. Each drives a different decision: ad sales, revenue and server capacity. *JioStar (IPL 2026 reach, July 2026; 30 crore subscriptions claimed in April 2025); ICC media release (semi-final concurrency, March 2026).*

The metrics that matter

Metric	What it means, in one line	Why a PM watches it
Paid subscribers, ARPU	People paying; revenue ÷ users for a period	Netflix stopped reporting subscribers in 2025 to focus on revenue
Churn	Share of paying users who stop in a month	At 5% a month, a service loses nearly half its base in a year
Watch time	Hours watched per user per week	The best early warning: people who stop watching cancel next
Time to first play	Seconds from opening the app to pressing play	Long browsing means “nothing to watch”, the top reason to cancel
Completion rate	Share of starts watched to the end	Tells you if a show held people, not only if its poster was good
Start time, rebuffering	Seconds to start; share of time spent frozen	The quality that viewers feel; 1% more freezing loses minutes watched
Ad fill, CPM	Share of ad slots sold; price per 1,000 ad views	Revenue = views × ads per view × CPM ÷ 1,000
Royalty per stream	What a music service pays rights owners per play	Music's biggest cost: it grows with every song played

Players and numbers

Player	What it is	Latest numbers (dated)
JioHotstar	Reliance and Disney's joint app: IPL and ICC cricket, TV serials, Disney, HBO	53 crore monthly users (Apr–Jun 2026); JioStar revenue ₹10,946 Cr that quarter
YouTube	Free video with ads, plus Shorts, Premium and Music	About 52 crore users in India (2026); 2.7 bn worldwide
Netflix	Global series and films; ₹149–649 a month in India, no ads plan here	Revenue \$12.6 bn in Apr–Jun 2026 (+13%); 2026 target margin 31.5%
Prime Video	Part of Amazon Prime (₹1,499 a year); ads by default since June 2025	Ad-free add-on ₹129 a month; also runs the free Amazon MX Player
Zee5, SonyLIV	TV network apps; regional shows, some sport	Smaller paying bases; often sold in bundles
Spotify	Music and podcasts; free with ads, or Premium	77.7 crore monthly users, 30 crore paying (Q2 2026); operating profit €655 mn
JioSaavn, Gaana	Indian music apps; strong in Hindi and regional songs	JioSaavn bundled with Jio; both fight YouTube Music for free listeners

Games	BGMI, casual puzzle games; real-money games now banned	Online Gaming Act in force from 1 October 2025
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Five turning points

Each turning point changed who pays, what they pay for, or how many people can watch at all.



Figure 6.6 • How Indian streaming changed in ten years. Cheap data created the audience; the rights split and a merger decided who owns cricket; a new law closed a whole category of games. *Company announcements; BCCI (June 2022); CCI and company filings (merger, completed 14 November 2024; JioHotstar launched 14 February 2025); Online Gaming Act 2025 (passed August, in force 1 October 2025).*

1. Cheap data (2016). After Jio launched, a gigabyte cost a few rupees instead of a few hundred. Video on the phone became an all-day habit for hundreds of millions of people. **2. Covid (2020).** Cinemas closed; films such as *Gulabo Sitabo* released straight on streaming, and many families paid for a plan for the first time. **3. Cricket goes digital first (2022–23).** The IPL's digital rights sold for almost as much as the TV rights. JioCinema then streamed IPL 2023 free to everyone, at scale, and taught the market that cricket on the phone could be paid for by ads. **4. One giant (2024–25).** Reliance's Viacom18 and Disney's Star India merged into JioStar, and JioHotstar launched in February 2025 with almost all big cricket, Disney, HBO content and TV serials in one app. **5. Real-money games banned (2025).** The Online Gaming Act banned games played for money, fantasy cricket included, from 1 October 2025. Dream11 and others turned to free games; ad spending on them vanished.

Five mechanics that aren't obvious

1. Sport buys sign-ups; series keep them. People join for the IPL and leave when it ends (Figure 6.7). A service that can't give them a reason to stay rents its audience for two months a year. **2. Churn and return.** Many viewers now subscribe, watch one thing and cancel, then come back for the next big show. The monthly count looks steady while different people fill it. Services fight this with annual plans, bundles and a steady flow of releases.

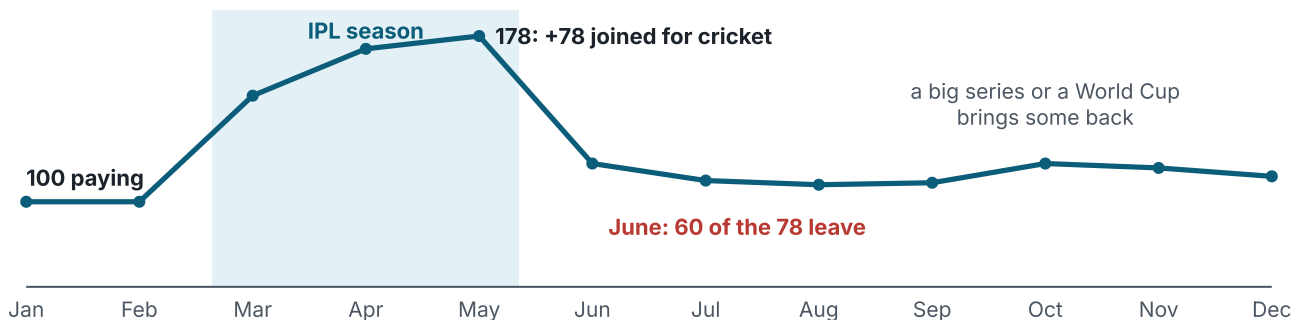


Figure 6.7 • Churn and return around the IPL. The paying base jumps for cricket and falls back in June. The question for the product is how many of the 78 can be kept. *Illustrative; the seasonal shape is widely reported for Indian sports streaming.*

3. Music pays per play. A video service pays for a show once. A music service pays record labels and publishers for every song played: about two-thirds of what it earns (Figure 6.8). More listening means more cost, so music apps push podcasts and audiobooks, which cost less per hour.

Where ₹100 of a music subscription goes (Spotify's cost mix, Q2 2026, rounded)

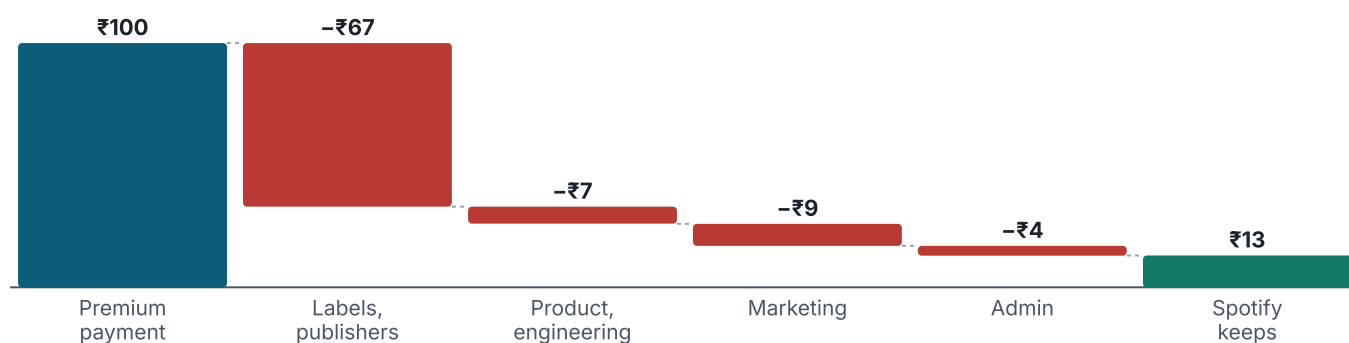


Figure 6.8 · Where a music subscription goes. Rights owners take about ₹67 of every ₹100. Spotify only turned a steady profit in 2024, after years of losses. *Spotify Q2 2026 results (gross margin 33.4%, operating income €655 mn on €4.8 bn revenue), shares rounded.*

4. A few hits, a long tail. A handful of titles take most viewing in any week; thousands of others are watched a little. The hits bring people in, the tail gives them something to do between hits, and recommendations connect the two. **5. Piracy is the free rival.** A popular film appears on pirate sites within hours of release, and a match is restreamed on Telegram. Services add invisible watermarks to trace leaks and, above all, keep prices low enough that paying is easier than hunting for a pirated copy.

6. A show is judged by what each hour of viewing cost. Streaming services compare very different titles with one number: **cost per hour watched** (what the title cost ÷ total hours people spent watching it). A ₹100 crore big-budget series watched for 2 crore hours costs ₹100 Cr ÷ 2 Cr = ₹50 an hour. A ₹12 crore small-town comedy watched for 1.5 crore hours costs ₹8 an hour (Figure 6.9). The cheaper show did six times better on this measure, which is why services keep commissioning modest Indian series such as *Panchayat*. Two cautions: a big title may also bring in new subscribers, which hours don't show, and a cheap title may be watched mostly by people who would have stayed anyway.

title	cost	hours watched	cost per hour watched
Big-budget series	₹100 Cr	2 Cr hours	₹50
Hollywood film licence	₹30 Cr	1 Cr hours	₹30
Small-town comedy	₹12 Cr	1.5 Cr hours	₹8
IPL match (digital rights)	₹58 Cr	20 Cr hours	₹3

IPL row: one match, 20 crore viewers × about 1 hour each = 20 crore hours (illustrative).

Figure 6.9 · Cost per hour watched. The expensive series drew the most hours, but each hour cost six times as much as the small comedy's. Live cricket looks cheap per hour because so many people watch at once; its risk is the total, ₹58 crore a match whatever the ad market does. *Illustrative budgets and hours; the measure is standard in streaming content teams.*

IN THE INTERVIEW · QUESTIONS THIS SECTOR LOVES

- **“How do you keep IPL subscribers after the season?”** Learn what else they like during the season, line up the next reason to watch, offer a pause instead of a cancel.
- **“Should Netflix cut prices in India?”** Say which segment, and test what they would pay first; a new cheap plan can pull existing members down to it.
- **“Views fell 20%. Why?”** Split views into viewers × titles started × share finished, and find which part moved.
- **“How would you handle 6 crore people watching at once?”** Prepare before the match, cache near viewers, and plan what to switch off first.
- **“Is YouTube a competitor to Netflix?”** Yes, for time: both sell the same evening hours.

CASE 6.1 · IMPROVE HOTSTAR

Improvement

Medium

IIM B Sigma PM Casebook 2026–27, pp. 105–106

Try it first. Think of the last time you paid for a streaming app for one thing, a cricket season or one series. Did you keep paying afterwards? What would have kept you? Now structure “improve Hotstar to increase loyalty” out loud for five minutes.

THE QUESTION

Can you walk me through how you'd improve the experience of an OTT platform like Hotstar?

A Before diving into the solutions, I'd like to clarify a few things to better scope the problem. What is the state of our company? What is the goal of the company at this point?

Q Let's assume the company is mature and has a significant urban user base. However, competition is intense, and we want to improve rider experience and increase loyalty.

A Understood. I'll start by outlining my high-level structure. First, I'll define relevant user personas basis engagement behavior. Then, I'll identify key pain points across their journey, followed by proposed solutions for the highest impact areas. Does that sound good?

Q Sure. Please Go ahead.

A I have come up with the following Personas:

1. The Casual Movie Critique: A casual movie critic who needs quick, fun, human style reviews before watching. He wants a strong community platform inside Hotstar to discuss movies, exchange opinions & fuel conversations, but hates boring, long text reviews.
2. The Social Reactor: A social reactor who watches what her friends & feed validate first. She wants meme like reactions, circle buzz signals and drops off quickly if content isn't socially hyped.
3. The Explorer: Arjun is an explorer who craves curated lanes to discover niche and hidden gems. He also wants to dive deeper into movie characters, learn about their background, and explore an actor's other movie.
4. Weekend Binger: Karan is a weekend-only watcher with limited time and high FOMO for top movies and sports moments. He wants instant picks and fast highlights
5. Casual Viewer: One who enjoys some casual sports, movies on a daily basis. Out of these segments, I want to address the Casual Movie Critique and Explorer, as their need for expressive reviews and deep discovery isn't satisfied well on OTT platforms. This also opens a opportunity for Hotstar to build a strong, differentiated

Q That's quite comprehensive. Can you list down the pain points of these viewers and your solutions?

A Pain Points:

Lack of an in app community platform for movie reviews, analysis, and discussion

No seamless way to watch with friends or engage through shared community moments

Limited actor/character discovery: no easy access to an actor's filmography or related suggestions

No support for user generated feedback formats like voice/video reactions and crowd ratings

No dashboard that shows total content watched so far or personalized watch-time insights

Features:

- 1). In app community forum (Reddit-like threads) for show/movie opinions, theories, and reviews
- 2) Shareable watch history card showing viewing time, patterns, and streaks for social credibility
- 3) Community Engagement Score visible on each show's page
- 4) Character lookup: actor bio, character arc, and filmography without exiting the movie
- 5) Group watch rooms with chat, and voting on next watch
- 6) Short show reels and user review clips indexed under the title for pre-watch discovery
- 7) User Reviews section

Q How would you prioritize among these solutions?

Feature	Reach	Impact	Confidence	Effort
Community Forum	H	H	H	H
Watch History Card	H	H	H	L
Group Watch	H	M	H	M
Short Reels / short Videos	H	M	M	M
User Reviews	M	M	H	L
Community Engagement score	H	H	M	M
Character Lookup	L	L	M	M

The casebook labels the candidate's next answer "Prioritization Using RICE framework".

A I evaluated each feature on reach, impact, confidence, and effort. To start with, I will prioritize launching the Community Forum, shareable Watch History Card, and Group Watch, as they collectively enable community trust, high engagement, and network effects for Hotstar.

Q Great, what would be some success metrics that you'd look at to see how well your improvement features are doing?

A

1. Churn Rate
2. DAU / MAU : in Forums
3. Average Watch time
4. Frequency of visit
5. Engagement per user

Q Great, we can close the case now.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked for the stage and the goal** first, and got a clear one: loyalty, in a crowded market.
- ✓ **Split users by how they behave**, not by age, which is the right kind of split for an engagement question.
- ✓ **Linked each pain point to a feature** and put the features in one table before choosing.
- ✗ **Never found where loyalty leaks.** Hotstar's biggest loss is the lakhs of people who join for cricket and leave when it ends. None of the seven features touches them.
- ✗ **Chose the smallest segments.** Movie critics and explorers are a few percent of users; loyalty is decided by the mass of casual and sports viewers.
- ✗ **"RICE" without numbers.** Letters (H, M, L) can't be multiplied. The forum is rated High effort yet ranked first; in RICE, high effort pulls a score *down*.
- ✗ **No goal metric.** Five metrics, none named as the one that shows loyalty (people still watching and paying months later).
- ✗ **Didn't check what exists.** Actor and character lookup already exists in India as Prime Video's X-Ray.

Scoping kit check: Confirm the thing ✓ · Goal ✓ **loyalty** · Who ✗ **which subscribers? (paid, bundled, cricket-only)** · Today and the trigger ✗ **why is loyalty low now?** · Edges ✗ **India only? which platform?** · Improvement add-on: which journey stage ✗. The costliest skip was "today and the trigger": one question about when people leave would have pointed straight at the June drop.

CHECK · THE CANDIDATE'S RANKING, REDONE WITH NUMBERS **DOESN'T HOLD UP**

RICE scores a feature as $\text{Reach} \times \text{Impact} \times \text{Confidence} \div \text{Effort}$. Here reach is subscribers affected in a quarter (in lakh), impact runs from 0.25 (tiny) to 3 (huge), and effort is team-months. All numbers are illustrative, but they are the kind you would defend in a room.

Feature	Reach (lakh)	Impact	Confidence	Effort	Score	Why
Community forum (candidate's #1)	20	0.5	50%	12	0.4	Few people post; needs moderation for spoilers and abuse
Group watch rooms	15	1	50%	8	0.9	Loved by a few; friends must be free at the same time
Character lookup	40	0.25	80%	4	2.0	Nice to have; Prime Video's X-Ray shows the idea works
Shareable watch-history card	100	0.25	80%	2	10	Cheap; brings a little free marketing
"After the IPL" watchlist (new)	300	1	60%	6	30	Aims at the biggest leak
Pause instead of cancel (new)	150	1	70%	3	35	Keeps the profile and history for the next season

With numbers, the forum drops to the bottom and two features aimed at post-IPL viewers go to the top. The letters hid this because "High" reach meant 20 lakh for one idea and 3 crore for another.

STRONGER ANSWER



Figure 6.10 · Six moves for "improve loyalty". Find where people leave before choosing who to design for. *Author's summary.*

A ① CLARIFY

By loyalty, do we mean people who keep paying and keep watching, month after month? Which subscribers: paying directly, or also those who get Hotstar with a Jio recharge? And is there a moment when loyalty drops? I'll assume India and paying subscribers.

Q Yes, renewals. Direct payers. Many people join in March for the IPL.

Why this move: "loyalty" can mean likes, visits or payments. Pinning it to renewals turns a vague brief into something you can count. The question about timing is what surfaced the IPL.

A ② FIND THE LEAK

Then I'd compare groups of people by the month they joined. Say 100 people join in January and 100 join in March for the IPL. By September, perhaps 65 of the January group still pay, against 28 of the March group (Figure 6.11). The March group isn't unhappy with the app. They came for one thing, and it ended.

Why this move: following each joining group separately shows a cliff that the total number of subscribers hides, because new joiners keep replacing the ones who leave.

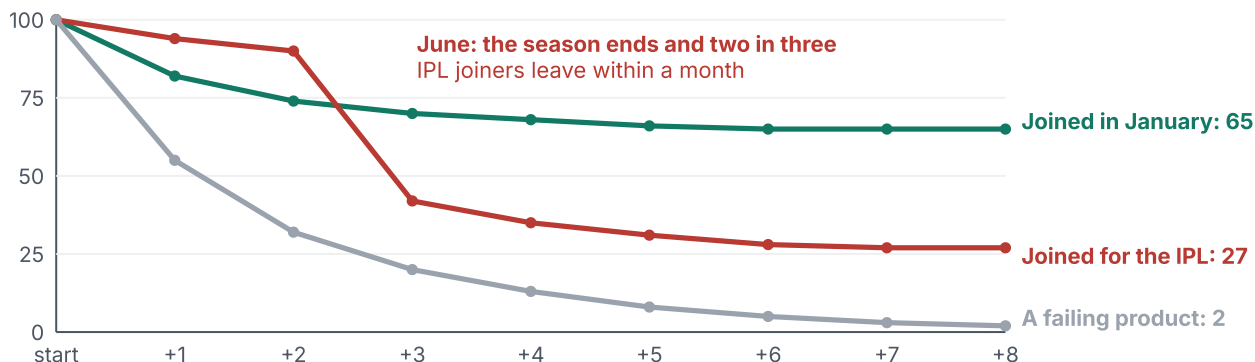


Figure 6.11 · Three groups of 100 people, followed month by month. The January group settles at about 65. The IPL group falls off a cliff in June. A third shape, falling towards zero, would mean the product itself is failing. Illustrative; shapes typical of Indian sports streaming.

A ③ SIZE

Illustrative: if 3 crore people subscribe for the IPL and 40% renew after June, that is 1.2 crore kept. Lifting renewal to 50% keeps 30 lakh more. On a ₹349 quarterly plan: $30,00,000 \times ₹349 =$ about **₹105 crore a quarter**, with no extra content cost, because the shows are already paid for.

Why this move: a rupee figure shows the interviewer the prize is large, and the "no extra content cost" part explains why keeping a subscriber is worth more than a new feature for a few enthusiasts.

A ④ WHO

The person to design for is Ravi, 26, a sales executive in Indore. During the IPL he opens the app four days in five, always for the match, sometimes for highlights. He has never started a serial or a film on Hotstar. On 1 June he has no reason to open it, so at renewal he doesn't.

Why this move: the candidate designed for critics and explorers, who already love content. Ravi is the one who leaves, so he is the one whose behaviour has to change.

A ⑤ BETS

Three ideas, the ones that score highest once RICE is done with numbers. **1. Pause, don't cancel:** when Ravi taps cancel, offer to pause for up to three months at ₹0, keeping his profile, and restart for the next tournament. **2. "After the IPL" watchlist:** during the season, learn what else he might like from small signals (he watches the Mumbai team, he skips ads with film trailers) and show a short list of shows at the final over of the final. **3. Cricket all year:** a calendar of the next matches (India tours, other leagues), with reminders he can switch on in one tap.

Why this move: each idea attacks the June cliff differently. The first makes returning easy, the second gives a reason to stay, the third makes cricket less seasonal.

A ⑥ MEASURE

Goal (North Star): share of IPL joiners who are still watching every week, 60 days after the final. **Inputs:** non-cricket titles started during the season; pause uptake; reminders switched on. **Safety check (guardrail):** revenue per subscriber, because a free pause could tempt loyal all-year payers to pause too. **Trade-off:** pushing serials during the IPL takes screen space from match ads, which pay the rights bill.

Why this move: counting people still watching, not only paying, catches the silent leavers early: people who stop watching cancel at the next renewal.

HOW THE ANSWER HANGS TOGETHER

Loyalty became renewals. Joining groups showed where renewals fail: after the IPL. The ₹105 crore sum sized the prize, Ravi made it human, and every bet aims at his June. The North Star (still watching 60 days later) and the guardrail (revenue per subscriber) measure exactly that cliff and the risk of the fix.

FRAMEWORK · RETENTION CURVES AND COHORTS

1 · The situation. Hotstar's total paying base looks healthy in April. But the people paying in April are not the people who paid last September. To see who stays, you have to follow each joining group on its own.

2 · An everyday example. A gym on 1 January. Sixty people join in the New Year rush. By March, twenty still come. In June, twelve are left, and those twelve keep coming for years. Meanwhile the owner sees 200 members on the register and feels fine, because each month's new joiners hide those who quietly stopped.

	The gym	Hotstar
A cohort	Everyone who joined in the January rush	Everyone who subscribed in IPL week one
What you track	How many still come in months 1, 2, 3...	How many still watch and pay in months 1, 2, 3...
Healthy shape	Falls, then flattens: a core of regulars	IPL joiners who stay for serials and films
Unhealthy shape	Keeps falling towards zero	A cliff in June that never flattens
The average trap	200 members on the register hides the leak	The April total hides last season's leavers

3 · The framework. A **cohort** is a group of users who started in the same period, such as everyone who subscribed in a given week. A **retention curve** plots the share of one cohort still active after 1, 2, 3... periods. Three shapes matter. A curve that **flattens** means the product has found people it keeps. A curve that **falls towards zero** means nobody stays: fix the product before buying growth. A curve with a **cliff** means something outside the product ended, like the cricket season. Put cohorts in rows and months across, and colour the cells: the table (Figure 6.12) shows at a glance whether newer cohorts keep people better than older ones.

Using it, step by step (on this case):

1. Choose what “retained” means: here, watched at least once in the week, not merely paying.
2. Group users by the week or month they joined, and mark the ones who joined for an event.
3. Draw one curve per group and name its shape: flattening, falling to zero, or cliff.
4. Compare a cohort with the same cohort a year earlier (IPL 2026 joiners against IPL 2025 joiners), never with a different season.
5. Tie each fix to a point on the curve: the pause and the watchlist aim at the June cliff.

COMMON SLIP · comparing the IPL cohort with the January cohort and calling IPL joiners “low quality”. They behave differently for a known reason; compare like with like.

Share of each joining group still watching weekly (%)

	start	month 1	month 2	month 3	month 4	month 5
Jan 2025	100	80	72	69	67	66
Mar 2025 (IPL)	100	93	90	40	33	30
Jul 2025	100	78	70	66	64	63
Jan 2026	100	83	76	72	70	69
Mar 2026 (IPL)	100	94	91	55	48	45

Red outline: June, when the season ends. 2026 IPL joiners: 55 kept, against 40 a year earlier.

Figure 6.12 · A cohort table. Each row is a joining month, each column is months since joining; darker means more of the group still watching. The IPL rows fade in June; the 2026 IPL row fades less than the 2025 one, which is what a fix should look like. *Illustrative.*

4 · When it misleads. Bundled users can make a curve look flat. A Jio recharge renews JioHotstar automatically, so a cohort of bundled users “stays” even if nobody opens the app. Count watching, not paying. Small cohorts also mislead: a test campaign that brought in 200 people can swing ten points either way by chance.

5 · In the interview. “Before choosing features, I’d look at retention by joining month; if the IPL cohort falls off a cliff in June, that is the loyalty problem to solve.”

CASE 6.2 · PRIME VIDEO VIEWS FALL 20%

Root cause

Hard IIM B Sigma PM Casebook 2026–27, pp. 80–81

Try it first. A “view” is counted when someone watches 70% of a title. Views per day have fallen 20% over a month, everywhere in India. Write down every way that number could fall without a single person cancelling. Five minutes.

THE QUESTION

The video views on Prime Video have gone down. What should we do? (Microsoft)

A Sure. Before I begin, I would like to ask some clarifying questions. What counts as a video view on Prime?

Q A video is counted as been viewed if at least 70% of the video is watched by the user. Views per day is the metric affected.

- A** Okay. How much has the decline been? For how long have we seen this issue? Has the decrease been gradual or sudden?
- Q** We have seen a 20% decrease. This has been gradual, over the last month.
- A** Is this issue prevalent across website and mobile apps on Android and iOS?
- Q** Yes, we are noticing this across all platforms, OS versions and devices.
- A** Is this issue happening in a particular geographical region in India? Or to a specific demographic group?
- Q** No. This is happening across India for all demographic segments.
- A** Understood. I will just reiterate the problem at hand. Video views on Prime Video is down by 20%. This has been gradual over the last month. We are seeing this issue on both website and mobile apps for all geographies and user segments in India. Can I proceed with this?
- Q** Yes. Please go ahead.
- A** Alright. I would first analyse some external factors that might have impacted our users. For internal factors, I would understand the journey of our user while watching videos on our platform. Does this seem okay to you?
- Q** The structure looks good. Please go ahead and list all the external and internal factors which you can think of.
- A** Sure! I will go over the external factors. Is there any major event going on currently which is being broadcasted? Something like the IPL or any football championships? This would have resulted in people shifting over there and spending less time on our application.
- Q** No. Nothing of that sort is happening.
- A** What about a competitor related change? This change could be due to the entrance of a new competitor or some changes introduced by an existing one. If the new entrant is a big player, users could be moving to them due to their reputation or some exclusive content. For an existing competitor, they could have introduced some discounts, new features or content to lure users.
- Q** Can you think of some other factors?
- A** Anything related to the government? Has any new legislation or regulation been introduced in India in this domain?
- Q** Nope. Nothing from the government side.
- A** Has there been some bad PR? The company's name being dragged through a scam or scandal.
- Q** No, nothing like that.
- A** Has this drop occurred earlier as well? Around the same time last year? This is just to make sure it is not a seasonal change.
- Q** No. This has happened for the first time.
- A** Okay. Since none of the external factors have affected, I am shifting to internal factors. Has there been any new app release or server deployment around the time when the decline started?
- Q** If it were a new release or deployment, wouldn't the change then have been sudden and not a gradual one?
- A** If this new release was rolled out gradually, its adverse impact would be gradual too.
- Q** Well pointed out. All our new releases and deployments are stable.
- A** If all these do not seem to be the issue I would go through the user journey and look at metrics at each step of the journey to see where things start to go wrong.
- Q** Please go ahead.

The candidate drew the user journey here: Login or Sign up → Recommendations or Search → Watch video. It is redrawn from the original in Figure 6.13.

- A** Here is the user journey for a typical user using our platform. I would list a few metrics at each stage of the journey and ask whether they have deviated significantly from the baseline over the last month.
- Q** No need. I think the diagnosis has been quite comprehensive. In this case, as you had mentioned earlier, there is a new application which is offering free content. People are spending their time over there. Now give me some recommendations to address this issue.
- A** Sure. The value proposition that people who migrated are looking for is value for money. This is understandable for the Indian market where people are price sensitive. So, we need to come up with an innovative way of pricing such as sachet pricing.
- Q** How would this sachet pricing work? Take the example of a series. How would it work?

A Okay. Sachet pricing would be for the whole movie and per season for a TV show. For illustration let us suppose we take the series Lupin. It has 5 episodes, and it has a new season which would get released next quarter. I would recommend that we price Lupin's entire season 1 and not do it per episode. Enabling payment per episode would mean that the users would need to pay again and again. This can result in a lot of dropouts in the user journey. Doing it season wise makes a lot of sense. Lupin season 1 ended on a cliff-hanger so people would be eager to comeback and pay for the next season when it arrives.

Coming to putting the actual number to the price, it would again depend on the type of content. If more people are interested in particular content, it could be priced high. But the price should not be so dynamic that it changes every few hours or days. This would discourage the viewers from paying if they would see the price rising for the content. Or they might be anticipating that the price might go down in the future.

Q Okay. This was quite interesting. The approach was very thorough. Thank you so much.

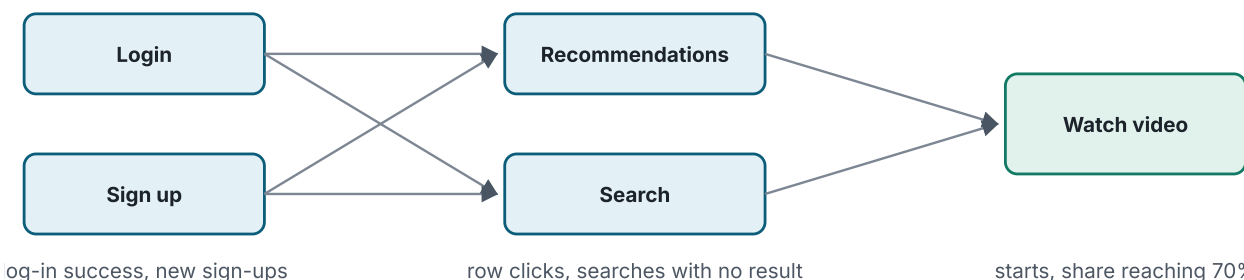


Figure 6.13 · The candidate's user journey, redrawn from the original. Two ways in, two ways to find a title, one way to watch. The numbers under each step are what the candidate proposed to check, added here as examples. *Redrawn from the original, IIM B Sigma PM Casebook 2026–27, p. 81; example metrics added by the author.*

WHAT WORKED · WHAT WAS MISSING

- ✓ **Excellent clarifying questions:** the exact definition of a view, how big, since when, sudden or gradual, which platforms, which regions and groups.
- ✓ **Played the problem back** in one sentence before starting.
- ✓ **A sharp answer on releases:** a change rolled out slowly would cause a slow fall.
- ✗ **Didn't use the definition.** Views = viewers × titles started × share watched to 70%. A 20% fall can hide in any of the three, and two of them don't need anyone to leave.
- ✗ **External checks were guesses, not data.** The interviewer had to reveal the cause. Splitting light and heavy viewers would have found it.
- ✗ **The fix doesn't match the cause.** People left for a *free* app. Paying per season is still paying.
- ✗ **Pay-per-title already exists.** Prime Video's rental store has sold single films in India since 2022.

Scoping kit check: Confirm the thing ✓ · Exact definition ✓ · Size and timing ✓ · Where it shows ✓ **platform, region, group** · Recent changes ✓ · Is the data trustworthy ✗ · Split of the metric ✗. A model opening. The one skipped question, “has the way we count a view changed?”, is worth asking whenever a fall is smooth and everywhere.

CHECK · “WE NEED AN INNOVATIVE WAY OF PRICING SUCH AS SACHET PRICING” **DOESN'T HOLD UP**

Two problems. First, Amazon already sells single films in India through the Prime Video rental store (since October 2022) and offers cheaper Prime plans such as Prime Lite. Second, Amazon's real answer to free rivals was *free*: its ad-supported miniTV, merged with MX Player into Amazon MX Player in 2024, and ads on Prime Video itself from June 2025. A viewer who left for free content won't come back to pay for a season of *Lupin*.

KEY IDEA · A SLOW FALL EVERYWHERE POINTS OUTSIDE THE CODE

A bug or a bad release usually hits one platform or one app version, and hits suddenly. A fall that is gradual and the same on every device, region and group usually means the world changed (a rival, a season, a habit) or what you offer changed (fewer new releases, longer shows), or the way you count changed. Check those three before opening the app's logs.

STRONGER ANSWER

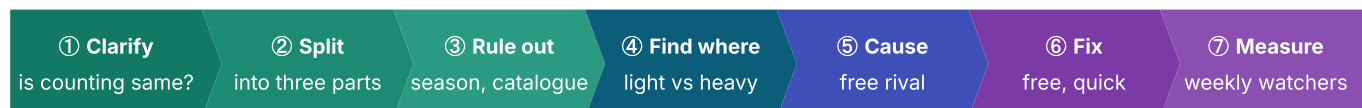
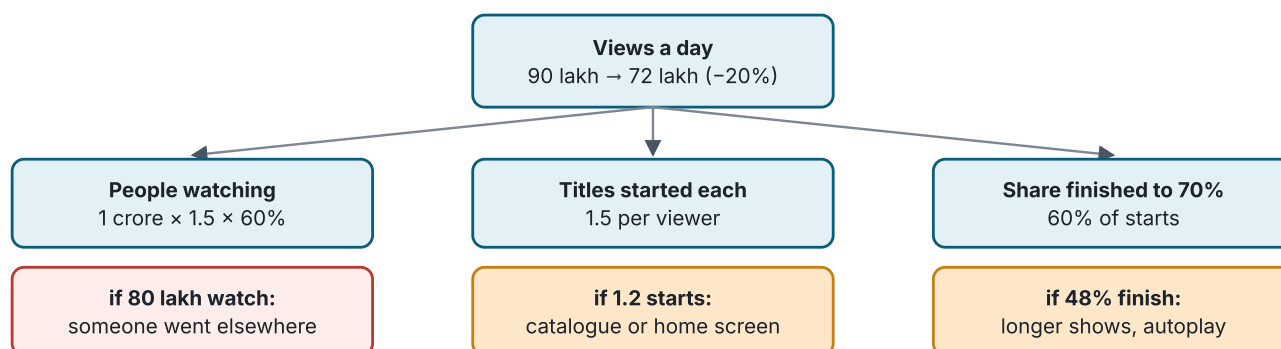


Figure 6.14 · Seven moves for a falling metric. The new move is ②: break the number into parts before guessing causes. *Author's summary.*

A ① **CLARIFY**

A view is 70% of a title watched, and views per day are down 20%, gradually, over a month, on every platform and in every region. Has anything changed in how we count a view, for example a new version of the player that logs progress differently?

- Q** No change in counting.
- Why this move: a counting change is the cheapest explanation to check and the most embarrassing one to miss. It takes one question.*
- A** ② **SPLIT THE NUMBER**
- Views per day = people watching that day × titles each person starts × share of starts watched to 70%. Illustrative: 1 crore viewers × 1.5 starts × 60% = 90 lakh views. A 20% fall means 72 lakh. That happens if viewers fall to 80 lakh, or starts fall to 1.2, or the finish rate falls to 48% (Figure 6.15). Which part moved?
- Q** Daily viewers are down about 15%. Starts per viewer dipped a little. The finish rate is flat.
- Why this move: each part has different causes. Fewer viewers means people went elsewhere; fewer starts means a weaker catalogue or home screen; a lower finish rate means longer shows or a player problem. Asking which part moved saves ten guesses.*



The interviewer's data: people watching -15%, starts slightly down, finish rate flat → people left.

Figure 6.15 · Three ways to lose 20% of views. Only the first needs people to leave. Longer episodes or autoplay trailers that count as starts would lower the finish rate with no one unhappy. *Illustrative numbers.*

- A** ③ **RULE OUT**
- Since viewers are down everywhere and gradually, it isn't a crash or one bad release. There was no IPL, no price change and no new rule. What about our own catalogue: were there fewer new releases this month than last year?
- Q** The release calendar was normal.
- Why this move: a thin month of releases is the most common internal cause of a slow fall in viewing, and it would look exactly like this.*
- A** ④ **FIND WHERE**
- Then which viewers are missing? I'd split by how often people watch. If the 15% is mostly light viewers, people who watched one or two evenings a week, they are the easiest to lose to something free and good enough.
- Q** Yes: mostly light viewers. They are watching films on a new free app.
- Why this move: the split by viewing habit turns "users left" into "one kind of user left, for one kind of reason", which is something you can design for.*
- A** ⑤ **CAUSE**
- For many light viewers, Prime Video came free with Prime, which they bought for delivery. They didn't cancel Prime; they just stopped opening Prime Video. So revenue hasn't fallen yet. The risk comes at renewal: if Prime's video part feels unused, Prime feels less worth ₹1,499.
- Why this move: naming who pays and why tells you how urgent this is and what to protect. Here it is Prime renewal, not a video subscription.*
- A** ⑥ **FIX**
- Meet free with free, and make choosing quick. **1.** Bring Amazon's free, ad-supported films into Prime Video's home screen for light viewers, so the free option is inside our app. **2.** A "tonight in under two hours" row: films and short series picked for people who watch once or twice a week. **3.** For new blockbusters, keep renting single titles; don't invent season passes that ask the same people to pay more.
- Why this move: every fix aims at the light viewer and the reason they left (free, easy), rather than at a pricing idea that asks them for money.*
- A** ⑦ **MEASURE**
- Goal (North Star):** Prime members who watch at least once a week. **Inputs:** days watched per light viewer; time from opening the app to pressing play. **Safety check (guardrail):** Prime renewal rate, and complaints about ads. **Trade-off:** more free, ad-supported titles bring light viewers back but can make paying for ad-free viewing feel less worth it.
- Why this move: the goal counts watchers, the unit that fell; the guardrail watches Prime renewal, the money that is really at risk.*

HOW THE ANSWER HANGS TOGETHER

Splitting views into viewers × starts × finish rate showed that people, not minutes, went missing. Splitting people by habit showed they were light viewers lured by free films. So the fixes are free and quick to choose, the goal counts weekly watchers, and the guardrail protects Prime renewal, which is where this problem would finally cost money.

Try it first. You have been watching Netflix on your cousin's account in another city for two years. Tomorrow it stops working. What is the most you would pay for your own plan? Would ₹99 to stay on your cousin's account change your answer? Five minutes.

THE QUESTION

The Indian OTT market is intensely competitive and price-sensitive. Netflix has already implemented price cuts and a mobile-only plan. Now, with the global rollout of its 'paid sharing' policy to curb password sharing, you, as a Product Manager at Netflix, need to refine the pricing and product strategy for India to continue growing the subscriber base and revenue.

- A** I understand that the objective here is twofold: continue to grow our subscriber base while also increasing revenue, specifically by converting password-sharers into paying customers.
- Q** Yes, that sounds fair. Go ahead.
- A** To structure my approach, I'll use the 5 Cs framework: Company, Customers, Competitors, Collaborators, and Context. I'll begin with what I know and then ask for a few details.
- Q** Sure, you may take your time and go ahead.
- A** 1. **Company (Netflix):** Our goals are shifting from pure subscriber growth to profitable growth, meaning Average Revenue Per User (ARPU) is now a key metric. The "paid sharing" feature is the critical global initiative to support this. Our core competencies remain our content engine and personalization algorithm.
2. **Customers:** The Indian market isn't monolithic. I see three key segments:
- **Mobile-First Youth:** Highly price-sensitive, needs a low-cost, mobile-only option.
 - **Family Household:** Values multiple screens and 4K quality for Smart TVs. They are less price-sensitive and prioritize convenience.
 - **Converted Sharer:** This is a new, important segment. They are being forced into a purchase decision and need an affordable, low-friction way to keep their profile and viewing history.
3. **Competitors:** I know the landscape is defined by bundling and low prices from major players like Amazon Prime Video, Jio Hotstar. To ground my recommendations in the current market, could you provide the latest pricing tiers for us and our key competitors?
- Q** Of course. Here are the current approximate monthly plans:
- Netflix: Mobile (₹149), Basic (₹199), Standard (₹499), Premium (₹649). Netflix doesn't offer annual plans.
 - Amazon Prime: Included with Prime membership at ₹299/month (or ₹1499/year), which also includes shopping and music.
 - Jio Hotstar: Has multiple tiers, with its premium ad-free plan at ₹299/month (or ₹1499/year).
- A** This data confirms my hypothesis: our premium plans are priced significantly higher than competitors, who leverage bundles (Amazon) and live sports (Jio Hotstar) as key value propositions. Our core value is our original content library. Competing with them on prices will neither justify our costs nor our value proposition. Moving on to cost-based pricing, it doesn't make sense to analyze it from a cost perspective, as the variable cost involved in serving every additional subscriber is close to null. This industry primarily comprises high fixed costs of obtaining content licenses, content production, IT infrastructure, etc., and hence shouldn't be looked at from a cost-plus pricing angle.
4. **Collaborators:** Telco operators like Jio and Airtel are key acquisition channels through bundling. OEM partnerships for pre-installation on devices also reduce friction.
5. **Context:** The password-sharing crackdown is the most significant trend. Additionally, the 5G rollout makes HD/4K plans more appealing, and the ubiquity of UPI simplifies payments.
- Q** That's a thorough analysis, especially highlighting the new "Converted Sharer" segment and the competitive pressures from bundled services. Based on this, what are your specific recommendations on the "paid sharing feature"?
- A** Based on that analysis, my recommendations are focused on successfully implementing paid sharing while refining our offerings for key segments.
1. **Optimize the "Paid Sharing" Feature for India:** This is our biggest opportunity to boost ARPU. The "Profile Transfer" feature must be flawless. I propose pricing the "add an extra member" slot at ₹99/month.
- a. **Rationale:** Our ad-free Basic plan is ₹199. At ₹99, the fee is less than half the cost of a new plan, making it a low-friction "yes" for someone forced to pay for the first time.

[The casebook's transcript ends here, after the first recommendation.]

WHAT WORKED · WHAT WAS MISSING

- ✓ **Restated a two-part goal:** more paying members and more revenue from former sharers.
- ✓ **Found the new segment,** the “converted sharer”, and described what that person needs: low price, keep my profile.
- ✓ **Asked for competitor prices** instead of guessing, and rightly set cost-plus pricing aside.

- ✗ **Recited a framework.** “5 Cs” filled time; only the customer segments were used in the recommendation.
- ✗ **Recommended something Netflix chose not to do in India.** When sharing rules reached India in July 2023, Netflix did not offer the paid “extra member” here, because prices had already been cut. The first question should have been “what do sharers in India get today?”
- ✗ **No test of what people would pay,** and no count of existing members who might move to a cheaper arrangement.
- ✗ **Stale competitor price.** JioHotstar’s Premium annual plan rose from ₹1,499 to ₹2,199 in January 2026. Prices move; say which date they are from.

Scoping kit check: Confirm the thing ✓ · Goal ✓ · Who ✓ **three segments** · Today and the trigger ✗ **what happens to sharers in India now?** · Edges ✗ **time frame**. Pricing add-ons: competitor prices ✓ · plan structure ✗ · cost floor ✓. Skipping “today” cost the whole recommendation: it rested on a product that doesn’t exist in India.

CHECK · “THE VARIABLE COST OF SERVING EVERY ADDITIONAL SUBSCRIBER IS CLOSE TO NULL” DIRECTIONALLY RIGHT

Content is fixed, so the big cost doesn’t grow with members. But some costs do: 18% GST comes off every rupee, payment fees take 1–2%, a telecom bundle gives the operator a cut, and a 4K viewer uses several times the bandwidth of a phone viewer. On a ₹149 plan, GST alone is about ₹23. “Small” is right; “close to null” is too strong, and it matters most on the cheapest plans.

STRONGER ANSWER



Figure 6.16 · Six moves for a pricing question. Know what exists today, then test what people will pay, then count who moves down. *Author’s summary.*

A ① CLARIFY

Two checks. Is the goal paying members, revenue, or both over the next year? And what happens today to someone in India using a relative’s account in another city: can they pay to stay, or must they get their own plan?

Q Both, over a year. In India they must get their own plan; they can move their profile across.

Why this move: whether an extra-member option exists decides the whole answer. One question avoids recommending a product that isn’t there.

A ② SEGMENTS

Four groups, each with a different price in mind. Young phone-first viewers: ₹149 or less. Families with a smart TV: will pay ₹499–649 for screens and picture quality. Former sharers: have never paid, so any price feels like a loss. And telecom customers: they get Netflix inside a recharge and never see its price.

Why this move: pricing is about what each group would pay, so the segments must be described by money and habit, not by age alone.

A ③ PRICE LADDER

Netflix costs more than its rivals at every step except the phone plan (Figure 6.17). JioHotstar’s phone plan is ₹79 a month; Prime costs about ₹125 a month on the annual fee and includes delivery. So Netflix shouldn’t race to the bottom. Its job is to make the first paid step, ₹149, an easy yes.

Why this move: laying the prices side by side shows where Netflix competes (phone plan) and where it charges for quality (TV plans), which keeps the fix narrow.

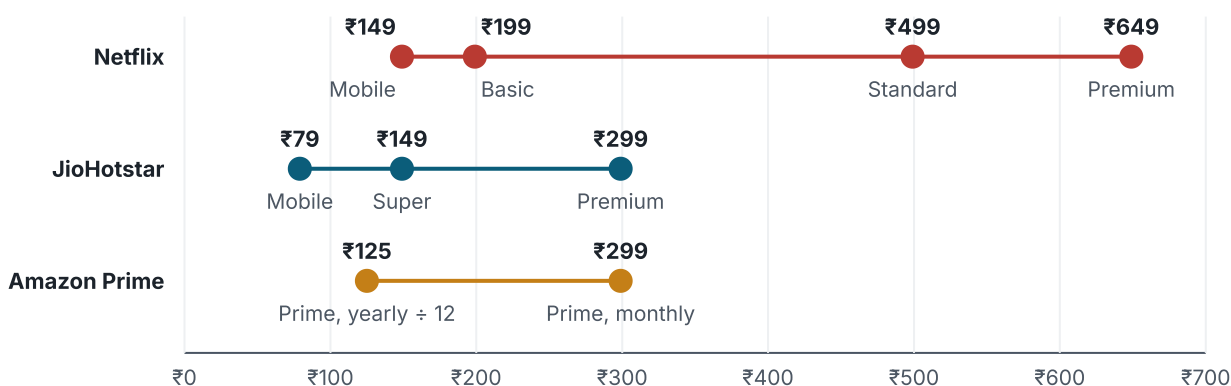


Figure 6.17 · Monthly prices in India, 2026. Netflix is the priciest at the top and matches rivals only at the phone plan. Prime’s price also buys free delivery, so it is not a like-for-like comparison. *Company websites (Netflix, JioHotstar, Amazon), 2026; annual plans divided by 12.*

A ④ TEST WHAT THEY'LL PAY

Before choosing a price, ask a sample of former sharers: would you pay ₹99? ₹149? ₹199? Illustrative answers: 70%, 45%, 25%. Revenue per 100 sharers: $70 \times ₹99 = ₹6,930$; $45 \times ₹149 = ₹6,705$; $25 \times ₹199 = ₹4,975$. The two lower prices earn almost the same (Figure 6.19).

Why this move: a quick survey turns "price-sensitive" into a curve, and the curve shows that going from ₹149 to ₹99 adds people but hardly any money.

A ⑤ COUNT WHO MOVES DOWN

Now compare the two options for 1 crore sharer households (illustrative). **Own plan at ₹149:** 25% sign up: 25 lakh $\times ₹149 = ₹37$ Cr a month. **Extra member at ₹99:** 45% pay: 45 lakh $\times ₹99 = ₹45$ Cr. But 10 lakh households that pay ₹499 for a second screen used by a sister elsewhere would switch to ₹199 + ₹99 = ₹298, losing ₹201 each: ₹20 Cr. Net ₹25 Cr. The own-plan route earns more (Figure 6.18).

Why this move: a cheaper option is sold to new people but also chosen by existing ones. Counting both is what separates a pricing answer from a guess, and here it explains Netflix's real choice.

A · "Get your own plan" at ₹149

1 crore sharer households
 $\times$ 25% buy their own plan
 $= 25 \text{ lakh} \times ₹149$
 $= ₹37$ Cr a month

B · Extra member at ₹99

1 crore sharer households
 $\times$ 45% pay to stay: $45 \text{ lakh} \times ₹99 = ₹45$ Cr
 – 10 lakh ₹499 households drop to ₹298:
 $10 \text{ lakh} \times ₹201 = ₹20$ Cr lost
 $= ₹25$ Cr a month

More people pay under B, but less money comes in: the cheaper option is also chosen by existing members.

Figure 6.18 · Two ways to charge former sharers. The ₹99 slot wins more people but pulls some ₹499 households down to ₹298. Illustrative numbers; the structure matches Netflix's decision not to sell extra members in India (July 2023).

A ⑥ RECOMMEND AND MEASURE

Keep "get your own plan", make ₹149 the landing plan with a one-tap profile transfer, and reach the most price-sensitive through Jio and Airtel bundles rather than a lower list price. Test an annual plan with a small discount to cut churn after big releases. **Goal (North Star):** revenue: paying members $\times$ revenue per member. **Inputs:** sharers converting within 30 days; profiles transferred; phone-to-TV upgrades. **Safety checks (guardrails):** cancellations and downgrades among existing members. **Trade-off:** a firm sharing rule gains revenue but annoys families spread across cities.

Why this move: the recommendation follows the numbers, and the guardrail watches the exact risk found in step ⑤: existing members trading down.

HOW THE ANSWER HANGS TOGETHER

The "today" question showed that no extra-member option exists in India. The survey showed ₹99 and ₹149 earn about the same from sharers, and counting existing members who would trade down made ₹149 the better price. Bundles reach the most price-sensitive without cutting the list price, and the guardrail watches downgrades, the risk the maths exposed.

FRAMEWORK · TIERED PRICING: GOOD, BETTER, BEST

1 · The situation. One Netflix catalogue is sold at four prices, from ₹149 to ₹649. The question is how to set those steps so each kind of viewer picks the plan meant for them, and nobody who would pay more slips down to a cheaper one.

2 · An everyday example. Indian Railways. The same train, the same journey, the same arrival time. A sleeper berth, a 3AC berth and a 2AC berth cost very different amounts. A student going home takes sleeper; a family takes 3AC; a manager on a company trip takes 2AC. You can't sit in 2AC with a sleeper ticket: the ticket checker enforces the fence.

	The train	Netflix India
Same core product	Same train and journey	Same shows and films
What changes by tier	Berth, air conditioning, crowding	Screens, picture quality, which devices
Cheapest tier is for	A student going home	A young viewer on a phone (₹149)
Top tier is for	A manager on company money	A family with a 4K TV (₹649)
The fence	The ticket checker	Phone plan plays only on phones, at 480p

3 · The framework. Tiered pricing sells one product at a few levels, usually three: good, better, best. Each level adds something a particular group values. A **fence** (a rule that stops people using a cheaper tier the way the dearer one is used) keeps people in the right tier. Charging different people different prices for nearly the same thing is **price discrimination**; the fences are device (phone-only plans), time (annual plans, student plans), bundle (inside a recharge) and country (Netflix costs far less in India than in the US). A **decoy** is a tier that exists mainly to make another look good: at a cinema, small popcorn ₹250, medium ₹380, large ₹400 makes large feel like a bargain. To set the steps, run

a **simple willingness-to-pay test**: ask a sample whether they would pay at four or five prices, and multiply the share who say yes by the price (Figure 6.19).

Using it, step by step (on this case):

1. List the groups and what each values most: price (phone viewers), screens and quality (families).
2. Give each tier one clear thing the tier below lacks, and one fence that stops trading down.
3. Test prices with the would-you-pay questions, and compute revenue at each price.
4. Count who moves: new buyers of a cheaper tier *and* existing customers who drop to it.
5. Launch the smallest change first, in one city or for one group, and watch downgrades.

COMMON SLIP · judging a new cheap tier only by the new customers it wins. The bill arrives from existing customers who move down to it.

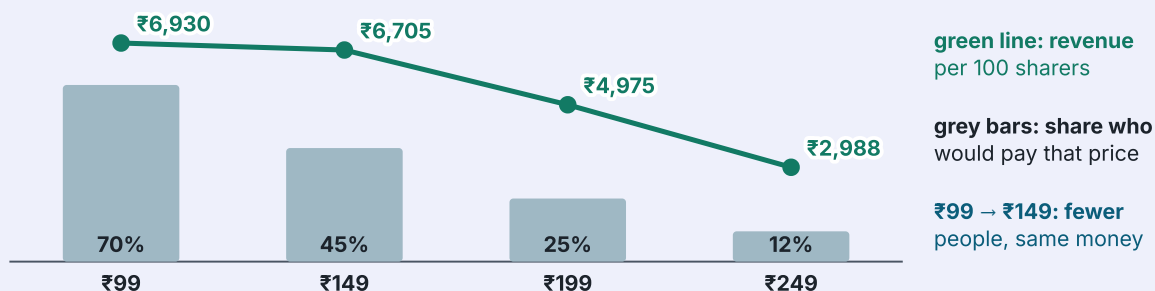


Figure 6.19 · A simple willingness-to-pay test. Bars show the share of former sharers who say they would pay; the line shows revenue per 100 of them. Revenue is almost flat from ₹99 to ₹149, then falls. *Illustrative survey answers.*

REAL-LIFE EXAMPLE · A FAMILY PLAN, WORKED

In the casebook's JioSaavn case, one person pays ₹99 a month and Spotify sells a six-person family plan for ₹179. Suppose JioSaavn launches a family plan at ₹149. A house where only the father paid now pays ₹149: +₹50, and three more people build listening habits, which makes the family less likely to cancel. But a house where four people each paid ₹99 (₹396) moves to one ₹149 plan: -₹247. So a family plan pays only where most families had one payer. The fence is "same home address", checked the way Spotify does, by location.

4 · When it misleads. Tiers that are too close make everyone pick the cheapest: if Basic at ₹199 differed from Standard at ₹499 only by one extra screen, few families would pay for Standard. And discrimination can feel unfair: in 1999, Coca-Cola's chief spoke of vending machines that would charge more on hot days, and the public anger forced the company to drop the idea. A fence that looks like punishment costs more than it earns.

5 · In the interview. "I'd keep three clear tiers, each with one fence, test what each group would pay, and count how many current customers would drop to any new cheaper option before launching it."

STREAMING, MUSIC & ENTERTAINMENT · GUESSTIMATES

Guesstimates: counting attention and fans

Media guesstimates come in two kinds. Money questions ("how much does YouTube earn a day?") are really about time: ads are sold by the minute of attention, not by the person. Audience questions ("how many people would use this app?") are about the difference between people who *could* use something and people who *will*, every month, when free rivals exist. The six cards still do the work.

1

Pin down the unit

Ad revenue or all revenue? Downloads, tried once, or used every month?

2

Pick the approach

For ad money, start from minutes watched; for users, from people who speak, own a phone, watch.

3

Draw the tree

People → watching today → minutes → ads per minute → price per ad.

4

Round numbers, one reason each

Ads per minute and price per 1,000 differ by screen: Shorts, phone, TV.

5

Check it another way

Against a published total (company results) and per person (does it sound like you?).

6

Give a range, and what moves it

Usually the price per 1,000 ads, or the share who would pick a new app over a free one.

KEY IDEA · AD MONEY = MINUTES × ADS PER MINUTE × PRICE PER AD

Counting "videos watched" undercounts ads: one 40-minute video can carry four mid-roll ads, and a 30-second Short often carries none. Start from minutes. For India, ₹ per 1,000 ad views is a small fraction of the US price, so the same minutes earn far less. That single fact explains most gaps between Indian and American media revenue.

Try it first. How many minutes of YouTube did you watch yesterday, and how many ads did you see? Were you on a phone or a TV? Build one day of YouTube's US ad revenue from that, for five minutes.

THE QUESTION

Estimate the revenue that YouTube makes in a day.

A Sure! I'll focus on YouTube's daily ad revenue. YouTube has several revenue streams ads, Premium, Music, TV and Super Chats but ads contribute the majority, so I'll concentrate on those. Should I estimate globally or for a specific market?

Q Let's estimate YouTube's daily ad revenue specifically in the United States.

A Perfect. Here's the approach I'll take: YouTube daily ad revenue = ad views × revenue per ad view, where

- ad views = total video views × % of videos carrying ads
- revenue per ad view = CPM ÷ 1,000

I'll estimate total views from the demand side how many videos people watch per day rather than the supply side, since YouTube can scale its infrastructure to meet demand, so capacity isn't the constraint. Does that sound fair?

Q Alright, that does sound like a plan. Let's go ahead.

A I'll take a US population of about 330 million, rounding to 300 million for cleaner segmentation, and estimate daily viewing behaviour by age group:

Age Group	% of Population	Viewership rate	Daily Users (Million)	Avg. Videos per day	Daily Views (Million)
0-5	10%	0%	0	0	0
6-12	10%	50%	15	5	75
13-24	20%	90%	54	20	1080
24-44	20%	85%	51	10	510
45-64	20%	60%	36	5	180
65+	20%	40%	24	3	72

A Total daily views ≈ 1,917 million ≈ 2 billion.

Not all views are monetized:

1. Some users have YouTube Premium (assume ~20% in US)
2. Some creators don't monetize
3. Some content has skippable or no ads

So, I'll assume 50% of views are monetized with ads.

Ad Views per Day ≈ 2B × 50% = 1B ad views

Q That sounds good, let's move forward.

A YouTube charges advertisers via CPM (Cost per 1000 Impressions). In the US, CPM varies by content and targeting. For this case, let's assume a CPM of \$20/1000 impressions. Revenue per ad view = \$0.02/ad view. Hence, daily revenue = 0.02 × 1B = \$20M.

For the year, we have \$20M × 365 = \$7.3 B

Q Anything else you would like to add?

A Google's annual ad revenue = \$200B. Assuming YT gives 10-15% of that, we have global YouTube revenues to be around \$20-30B. Hence, assuming USA gives \$7.3B is a fair estimate.

Q Okay, we can close the interview.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Scoped the unit** (ad revenue only) and asked which market.
- ✓ **Wrote the equation first**, and explained why demand, not server capacity, sets the number.
- ✓ **Removed views without ads** (Premium, creators who don't earn from ads) and checked against Google's total.

✗ **Counted videos, not minutes.** One ad per view undercounts long videos with mid-rolls and TV viewing, now YouTube's biggest screen in the US.

✗ **Overlapping age bands:** 13–24 and 24–44 both include 24-year-olds, and 0–5 at 0% ignores YouTube Kids.

✗ **The check was misread.** \$7.3 bn from the US out of \$20–30 bn worldwide would make the US only a quarter to a third of YouTube, when the US brings in about half of Alphabet's revenue. And YouTube's ads alone were \$36 bn in 2024.

Guesstimate kit check: 1 Unit ✓ · 2 Approach ✓ **demand side** · 3 Tree ✓ · 4 Numbers with reasons **X age rates unexplained** · 5 Check **X done, but it should have raised a flag** · 6 Range **X**. Card 5 again: the check is only useful if you let it change your answer.

CHECK · “ASSUMING USA GIVES \$7.3B IS A FAIR ESTIMATE” DOESN'T HOLD UP

Alphabet reported YouTube ad revenue of \$36.1 bn in 2024 and about \$40 bn in 2025. It doesn't split YouTube by country, but about half of Alphabet's revenue comes from the US. Half of \$40 bn is about \$20 bn a year, or $\$20 \text{ bn} \div 365 \approx$ **\$55 million a day**, more than twice the casebook's \$20 million.

STRONGER ANSWER



Figure 6.20 · The six cards, applied to ad revenue. Minutes (card 2) and a published total (card 5) do most of the work. *Author's summary.*

A 1 · UNIT

YouTube's ad revenue in the US on a typical day, including Shorts and viewing on TV screens; Premium and YouTube TV subscriptions are out.

Why this move: “revenue” could include subscriptions, which are large on their own. Saying what is in and out avoids an argument later.

A 2 · APPROACH

Start from minutes watched, because ads are placed by time: a long video carries several ads, a Short few or none.

Why this move: minutes match how the product actually earns, so the rates in the tree become ones you can defend.

A 3–4 · TREE AND NUMBERS

330 million Americans $\times$ 75% watch on a typical day $\approx$ 250 million viewers. $\times$ 60 minutes each = 15 billion minutes. $\times$ 85% not on Premium = 12.75 billion minutes with ads. $\times$ 1 ad every 4 minutes = 3.2 billion ad views. $\times$ \$15 per 1,000 (cheap on Shorts, dear on TV) = **about \$48 million a day** (Figure 6.21).

Why this move: each number has a reason: an hour is typical US viewing across phones and TVs, one ad per four minutes blends long videos and Shorts, and \$15 averages cheap and expensive screens.

A 5 · CHECK

Against the company: YouTube ads were about \$40 bn in 2025; if the US is roughly half, that is $\$20 \text{ bn} \div 365 \approx$ \$55 million a day. Per person: \$48 million $\div$ 250 million viewers $\approx$ 19 cents a day, about \$70 a year, which fits a market where Meta earns about \$68 a quarter per user across all its apps. Both checks land near \$50 million.

Why this move: a company total and a per-person figure attack the answer from two sides. When both agree, the estimate is safe.

A 6 · RANGE

\$30–75 million a day, most likely about \$50 million, or roughly \$18 bn a year. The swing factors are ads per minute and the price per 1,000, which rises as more viewing moves to TV screens.

Why this move: the range tells the interviewer which two assumptions to argue about, which is more useful than a single number.

HOW THE ANSWER HANGS TOGETHER

Scoping to ad revenue fixed the unit. Starting from minutes matched how YouTube sells ads, which is why this answer is more than twice the casebook's. The company total and the per-person check both point to about \$50 million a day, and the range names ads per minute and TV prices as what would move it.

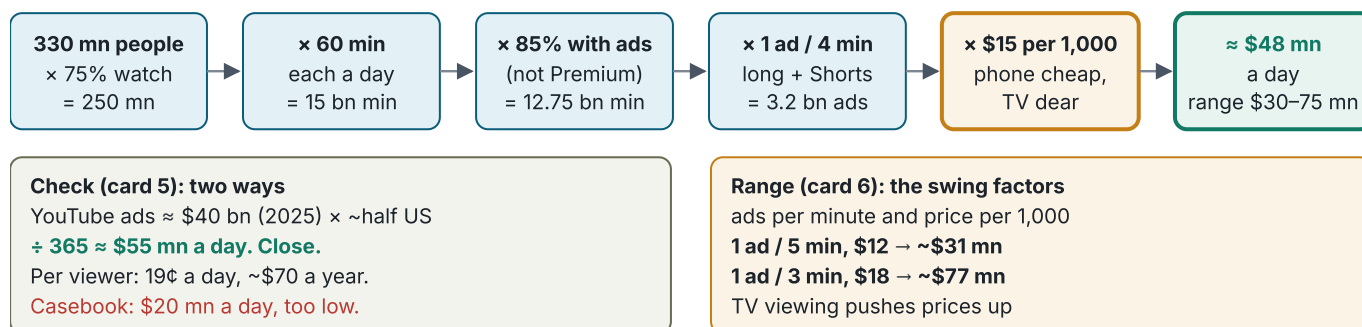


Figure 6.21 · YouTube's US ad revenue a day, as an equation tree. The orange box is the swing factor; the green box is the answer; the grey box is the check against Alphabet's reported figures. *Alphabet annual reports (YouTube ads \$36.1 bn in 2024; about \$40 bn in 2025; US about half of Alphabet's revenue); viewing, ad-load and price assumptions illustrative.*

THE REUSABLE TRICK

Ad revenue = viewers $\times$ minutes $\times$ ads per minute $\times$ price per ad. Check it against the company's annual figure divided by 365, and against revenue per person per day.

Try it first. Where do Bhojpuri speakers you know watch Bhojpuri films today, and do they pay? Estimate how many would use a new Bhojpuri-only app every month. Five minutes.

THE QUESTION

Estimate the number of potential users for a Bhojpuri-language short-movie app.

- A** I have a few clarifying questions. Are we estimating users in India only, or should I include the global Bhojpuri-speaking diaspora as well? And should I estimate everyone who speaks Bhojpuri, or narrow it down to those who could realistically use a smartphone app?
- Q** Good questions. Include the diaspora, and yes, narrow it down to realistic users.
- A** I'll estimate this in three steps: first the total Bhojpuri-speaking population (TAM), then the portion with smartphones and regular data access (SAM), and finally the realistic share who'd actually adopt this app (SOM). Is this approach fine?
- Q** Sounds like a comprehensive approach, please proceed.
- A** Starting with TAM, the total Bhojpuri-speaking population:
- Bihar population ~130 million → ~45% Bhojpuri-speaking → ~59 million
 - Eastern UP population ~240 million → ~20-25% in the Bhojpuri belt → ~50 million
 - Jharkhand border areas → ~5 million
 - Global diaspora (Nepal, Gulf countries, Mauritius, Fiji, Caribbean) → ~6-8 million
- Adding these up, TAM ≈ 120-125 million Bhojpuri speakers.
- Does this breakdown seem reasonable?
- Q** Yes, you may continue.
- A** Not all speakers are potential app users, so I'll apply two filters to narrow this to SAM: smartphone penetration and regular data usage.
- Smartphone penetration in the Bhojpuri belt: ~65%
 - Regular data users among them: ~75%
- $SAM = 125 \text{ million} \times 0.65 \times 0.75 \approx 61 \text{ million reachable users.}$
- Does this approach seem reasonable?
- Q** Yes, that's a fair filtering approach.
- A** Now, for SOM, I'll segment the reachable users by likelihood of actually trying the app:
- **High-likelihood adopters** urban migrant workers and youth: ~30% adoption
 - **Medium likelihood** – Tier 2-4 families: ~15% adoption
 - **Low likelihood** – NRI diaspora: ~10% adoption
- Weighting across these segments gives a blended expected adoption rate of roughly 20-25% initially.
- Q** That seems like a reasonable spread. Please go ahead and calculate the final number.
- A** Taking the midpoint of ~22%:
 $SOM = 61 \text{ million} \times 0.22 \approx \mathbf{13-14 \text{ million}}$
- Total number of potential users for a Bhojpuri Short-film app** = ~13-14 million.
- Q** This was a comprehensive estimate. Thank you.

[The casebook prints this case in two columns; the turns are set out here in reading order, in the casebook's words.]

WHAT WORKED · WHAT WAS MISSING

- ✓ **Two sharp clarifying questions:** India or the diaspora too, and all speakers or only realistic users.
- ✓ **A clean funnel** from speakers to people online to people likely to adopt, with the sums shown.
- ✓ **Split adoption by segment** rather than one guess for everyone.
- ✗ **"Users" never defined.** People who try it once, or people who use it every month? 1.3 crore triers and 1.3 crore monthly users are very different businesses.
- ✗ **No free rival in the maths.** Bhojpuri songs and films are already free on YouTube and Moj. A 22% adoption rate for a new app ignores that.
- ✗ **The blend isn't shown.** The 20–25% depends on how many of the 6.1 crore are youth, families or diaspora, and those shares were never given.

Guesstimate kit check: 1 Unit ✗ **"potential user" undefined** · 2 Approach ✓ **top-down funnel** · 3 Tree ✓ · 4 Numbers with reasons ✗ **adoption rates unexplained** · 5 Check ✗ · 6 Range ✓ **13–14 million, narrow**. Card 1 mattered most: the interviewer asked for

“potential users”, and the answer treated that as people who would adopt.

CHECK · “TAM ≈ 120–125 MILLION BHOJPURI SPEAKERS” **DIRECTIONALLY RIGHT**

The 2011 Census counted about **5.1 crore** people whose mother tongue is Bhojpuri; many more speak it but record Hindi. So 12 crore is a fair upper figure for people who understand Bhojpuri, and 5 crore a floor. The online filter (49%) is high for Bihar, which has among India’s fewest internet users per person (TRAI).

STRONGER ANSWER



Figure 6.22 · The six cards, applied to a new app’s audience. Defining the user (card 1) and the free rival (card 4) change the answer most. *Author’s summary.*

A 1 · UNIT

I’ll give two numbers: people who might try the app, and people who would use it every month once it is well known, say in its third year. Both include the diaspora.

Why this move: “potential users” is ambiguous; giving both numbers keeps the answer useful whichever was meant.

A 2 · APPROACH

Top-down from speakers: people who understand Bhojpuri → online → already watch short videos → would choose a Bhojpuri-only app over free options.

Why this move: the last step names the real competitor.

A 3–4 · TREE AND NUMBERS

About 9 crore people understand Bhojpuri (between the Census’s 5 crore and the candidate’s 12 crore). × 55% on-line = 5 crore. × 80% watch short videos every month = 4 crore. Of those, perhaps 25% would try a Bhojpuri app: **about 1 crore triers**. And 5–10% would use it every month despite YouTube and Moj: **20–40 lakh monthly users** (Figure 6.23).

Why this move: the gap between trying (25%) and staying (5–10%) is the free rival at work.

A 5 · CHECK

ShareChat and Moj together have about 15 crore monthly users across more than 15 Indian languages, roughly 1 crore per language on average. A single-language, single-format app reaching 20–40 lakh monthly users sits sensibly below that average.

Why this move: a real, many-language app gives a ceiling per language.

A 6 · RANGE

About 1 crore people might try it; **20–40 lakh would use it monthly**. The swing factor is the share who stay once the novelty wears off, which depends on having films you can’t find free elsewhere.

Why this move: it turns the estimate into a product insight: exclusive content decides the size.

HOW THE ANSWER HANGS TOGETHER

Defining the user split the answer into triers and monthly users. The Census gave a floor for speakers, the free rival explained the drop from trying to staying, and the ShareChat–Moj average capped it. The range points at exclusive content, the one lever that would make a Bhojpuri app bigger.

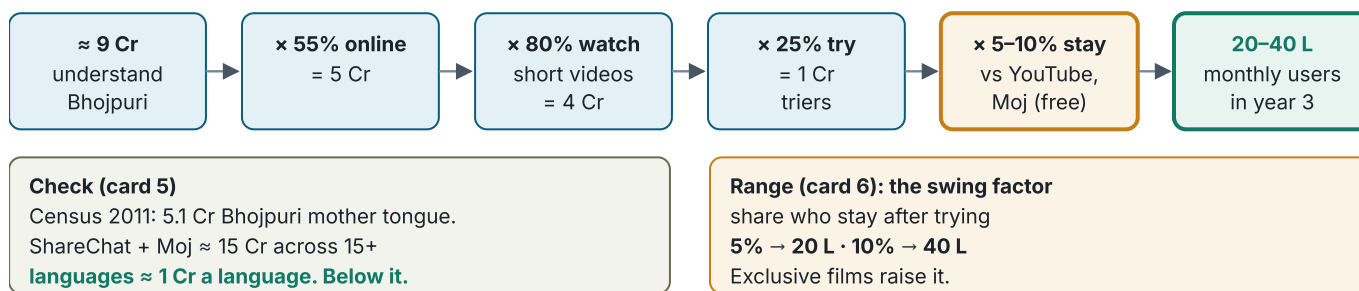


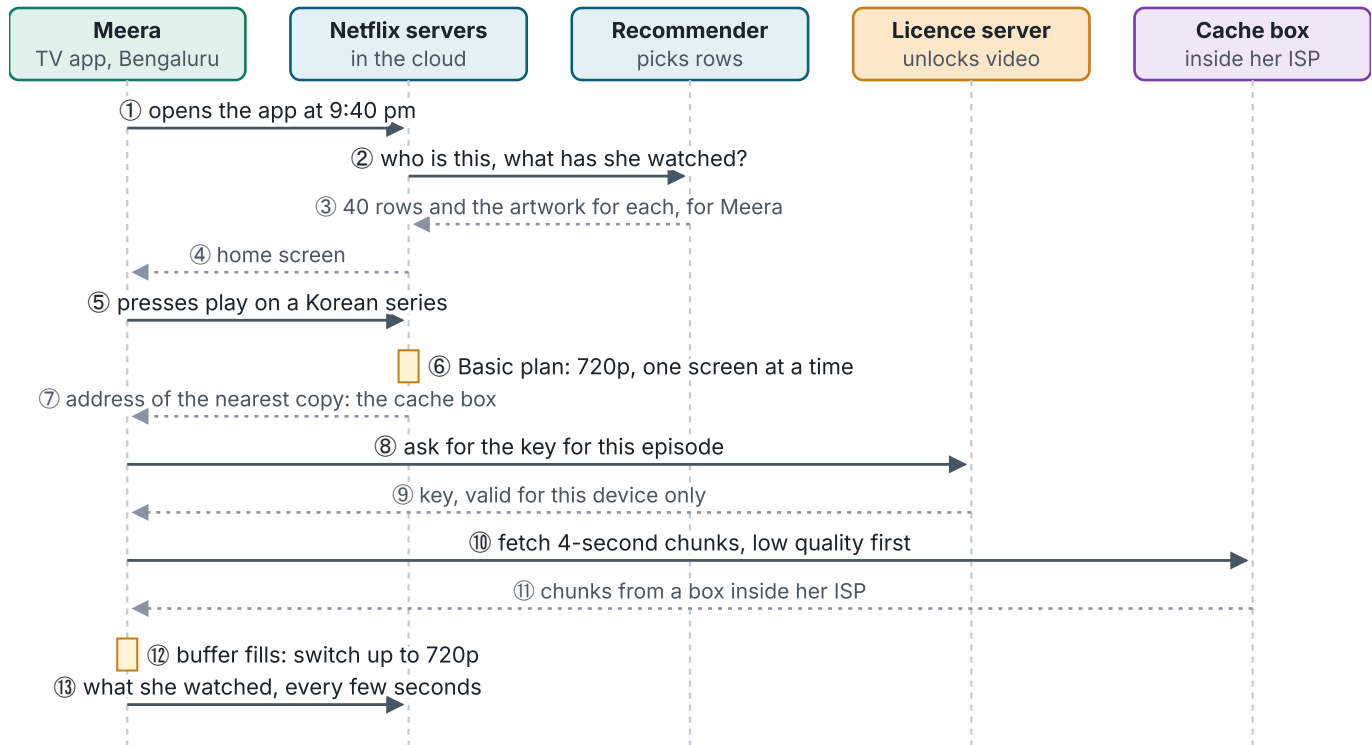
Figure 6.23 · A Bhojpuri app’s audience, as an equation tree. The orange box is the swing factor; the green box is the answer; the grey box is the check. *Census of India 2011 (Bhojpuri mother tongue, about 5.1 crore); ShareChat and Moj monthly users (company, 2026, about 15 crore); other shares illustrative.*

THE REUSABLE TRICK

For any new app: people who could use it → online → already do this activity → try it → stay against the free option. Check against a similar app’s users per language or per city.

Netflix: always something good to watch, found in a minute

Walkthrough (illustrative persona). Meera, 29, a software tester in Bengaluru, pays ₹199 a month for Netflix Basic and watches on her TV. At 9:40 pm she opens the app. The home screen is built for her alone: a row of Korean dramas because she finished one last week, a “Top 10 in India today” row, and a poster for a thriller that shows its female lead, because Meera tends to click on those. She presses play. The picture appears in about two seconds, a little soft, and sharpens within ten. When the episode ends, the next one starts in five seconds unless she stops it. She watches three (Figure 6.24).



Nothing comes from America: the episode was copied to the box in her ISP overnight.

Figure 6.24 · Behind Meera's evening. The home screen is chosen for her (2–4); the licence key (8–9) stops copying; the video itself comes from a Netflix box inside her own internet provider (10–11), which is why it starts fast. *Simplified from Netflix's technology blog and Open Connect documentation (2016–26); timings illustrative.*

Feature	User problem	Why it's designed this way	Metric it moves
Personal rows and artwork	"Too much choice; nothing to watch"	Each row, and even each poster image, is chosen for the viewer; the aim is to press play in under a minute	Time to first play; hours per member
Profiles	"My son's cartoons ruin my suggestions"	A separate history for each person in the house	Quality of recommendations
Autoplay next episode	"I want to keep going without reaching for the remote"	Five-second countdown; can be switched off	Episodes per session; series completed
Downloads	"The internet on my train is patchy"	Save episodes on the phone at home, watch offline	Mobile-plan retention
Phone-only plan (₹149)	"I can't pay ₹499"	Cheap, but only on phones at 480p, so families still buy TV plans	New members in India
Household rule, profile transfer	Netflix: millions watching free on others' accounts	Accounts tied to one home; a sharer can move their profile to a new account	Paying households

Core loop: why people come back, and why they stay

Each evening follows a small loop: open, see something picked for you, press play, finish, see the next suggestion. The bigger loop is the business (Figure 6.25). More members bring more money; more money pays for more shows in more languages; more hits bring more viewing and more data about what people like; better picks mean less browsing, so fewer people cancel, and the base grows again. The loop only works at a size where a ₹200-crore series is shared by crores of members.

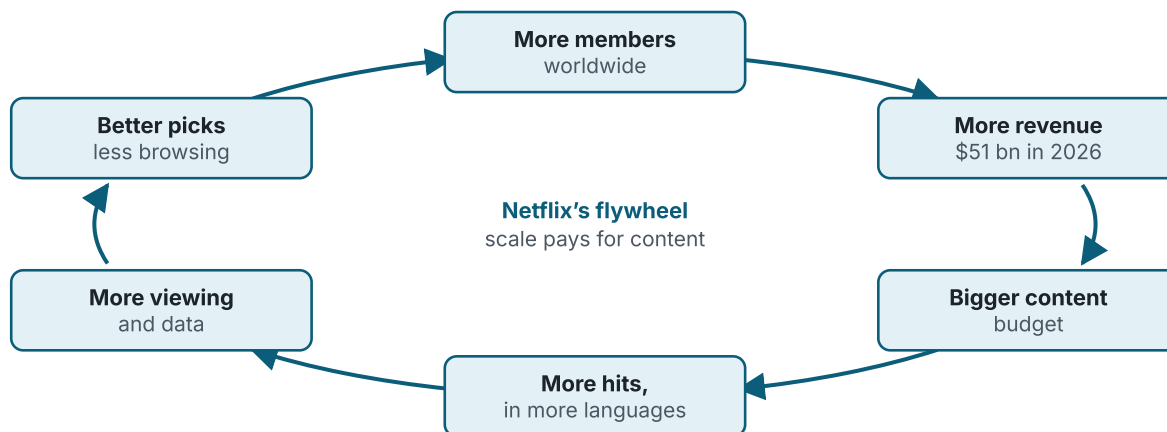


Figure 6.25 • Netflix's flywheel. Each step makes the next one easier. Scale is the moat: a new rival must pay similar content costs with far fewer members. *Author's diagram; revenue guidance from Netflix's Q2 2026 letter.*

Business model, with numbers

Netflix earns almost entirely from subscriptions, plus a fast-growing ads plan in about a dozen countries (not India). It expects revenue of about **\$51 bn in 2026**, with an operating margin of 31.5%; April–June 2026 revenue was \$12.6 bn, up 13%. Advertising should roughly double, to about \$3 bn, in 2026. Content is the largest cost, spread over the years each show is watched. In 2025 Netflix stopped reporting member numbers, to point investors at revenue and profit instead. In India it chose low prices over ads: a ₹149 phone plan and bundles with Jio and Airtel.

Moat, in plain words. It can spend more on shows than almost anyone because the cost is shared by hundreds of millions of homes. It owns its big originals worldwide, so *Squid Game* exists only here. And years of viewing data make its picks better than a newcomer's.

NORTH STAR Hours of viewing per member that people rate as satisfying.	INPUTS Time to first play · share of members starting a new release in 30 days · series completion · ads-plan reach.	GUARDRAILS Monthly cancellations · cancellations after a price rise · ad load on the ads plan.
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2025–26 moves. Live events (sport and specials) to create “watch it tonight” moments; games on TV; video podcasts; short vertical clips in the phone app from August 2026; and the password-sharing rule, now in every major market.

STRENGTHS

- ✓ Global scale spreads content cost
- ✓ Recommendations that cut browsing
- ✓ Very profitable; can outspend rivals

WEAKNESSES

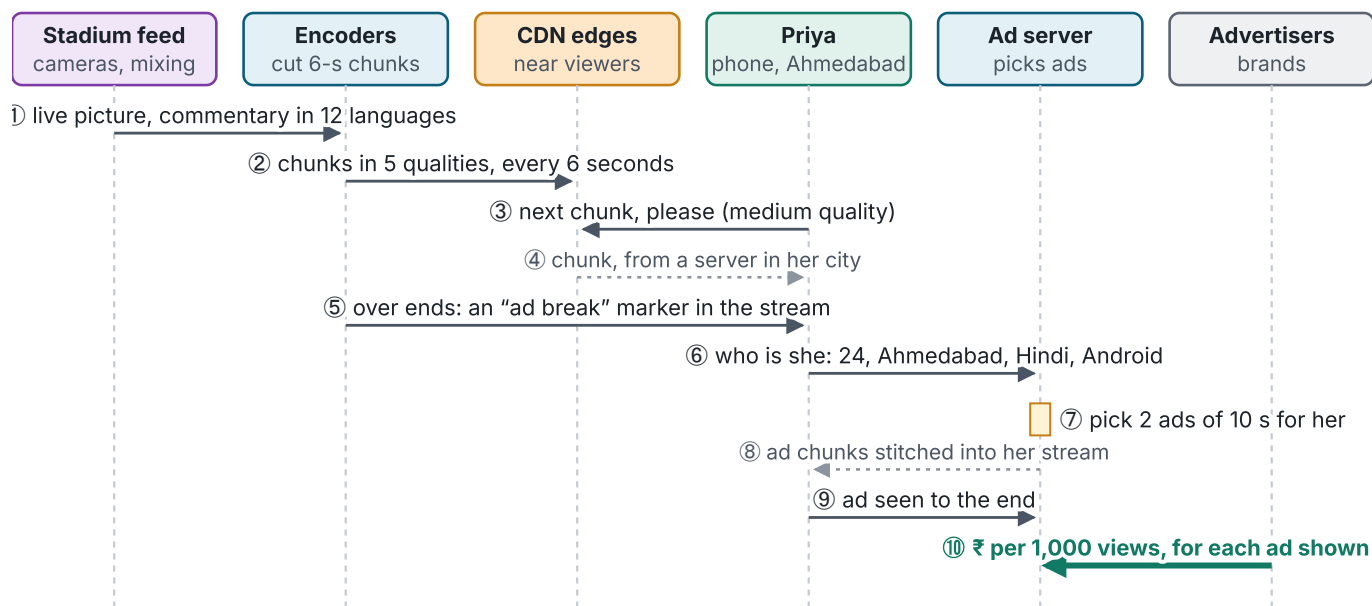
- ✗ Little live sport, the biggest draw in India
- ✗ Priciest TV plans in India
- ✗ Slow growth outside cities

What I'd build next

1. During the IPL, non-cricket viewers in a house fight for the TV → “match-length” picks: films under two hours and one-night stories, promoted at 7:30 pm on phones → hours per member in April–May; cancellations in May.
2. Growth outside big cities is slow → more Tamil, Telugu and Malayalam originals, with good subtitles and dubbing into eight languages → sign-ups outside metros; share of viewing in Indian languages.
3. People join for one show and leave → a small discount for paying for six months, offered on the day a big series ends → share of members on longer plans; churn in the month after big releases.

JioHotstar: cricket for everyone, paid for by ads

Walkthrough. Back to Priya on the hostel steps in Ahmedabad on final night. She pays ₹79 a month for the phone plan, but most of her friends watch on a plan bundled with their Jio recharge. She picks Hindi commentary. The picture starts blurry and sharpens. At the end of the over, two 10-second ads play: a phone brand and a biscuit brand, in Hindi. Her friend in Chennai, watching the same ball, sees two different ads in Tamil. Nobody chose those ads by hand: a computer picked them for each of the crores of phones watching (Figure 6.26).



Crores of people see different ads in the same 30-second break.

Figure 6.26 • Behind one over-break. The video is the same for everyone; the ads are not. A marker in the stream (⑤) tells each phone to ask for its own ads (⑥–⑧), so one 30-second break is sold many times over. *Simplified from how server-side ad insertion generally works; JioHotstar's internal design is not public; numbers illustrative.*

Feature	User problem	Why it's designed this way	Metric it moves
Commentary in 12 languages	"I want the match in my language"	Separate audio for Hindi, Tamil, Telugu, Bhojpuri and more, chosen in one tap	Reach outside Hindi states
Cheap plans and bundles	"I won't pay ₹500 a month"	Phone plan from ₹79 a month; included in many Jio recharges	Paid subscriptions
Vertical video, key moments	"I joined late" / "I'm on my phone"	Watch holding the phone upright; jump to wickets and sixes without leaving live	Minutes per viewer
Serials, Disney, HBO, regional films	"Something for the rest of the family"	One app for sport and the evening serial	Watch time outside cricket
Tadka (short videos)	"I have five minutes"	Vertical micro-dramas and clips	Daily users between matches

Business model, with numbers. JioStar earns from ads (sport above all), subscriptions (many through recharges) and TV channels. In April–June 2026, the IPL quarter, its revenue was **₹10,946 crore**, up 14%, and JioHotstar averaged 53 crore monthly users. Prices rose in January 2026: phone ₹79 a month, Super ₹149, Premium ₹299. The biggest cost is rights: IPL, ICC events and studio libraries, paid for whether or not the ads sell.

Moat, in plain words. Almost all top cricket in one app; Jio's hundreds of millions of phone customers as a sales channel; and the engineering and ad-sales machine to serve crores at once. The risk is the same as the moat: rights cost a fortune, and a weak ad year doesn't lower them.

NORTH STAR Monthly viewers who also watch something other than cricket.	INPUTS Peak concurrency · ad slots sold at peak · cricket-to-serial viewers · paid plans outside bundles.	GUARDRAILS Buffering at peak · complaints about too many ads · cancellations after the IPL.
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COMPARISON • PRIME VIDEO: THE SHOW THAT SELLS THE SHIPPING

Prime Video earns in a different way. For most Indians it comes inside Amazon Prime (₹1,499 a year), which also gives free, fast delivery. Its job is to make Prime feel worth renewing, so that members keep shopping on Amazon (Figure 6.27). Since 17 June 2025 it shows ads by default in India; removing them costs ₹129 a month. Its pitch to advertisers is not JioHotstar's mass reach but ads matched to real shopping data, with sales measured on Amazon. For free viewers, Amazon runs Amazon MX Player, with ads.

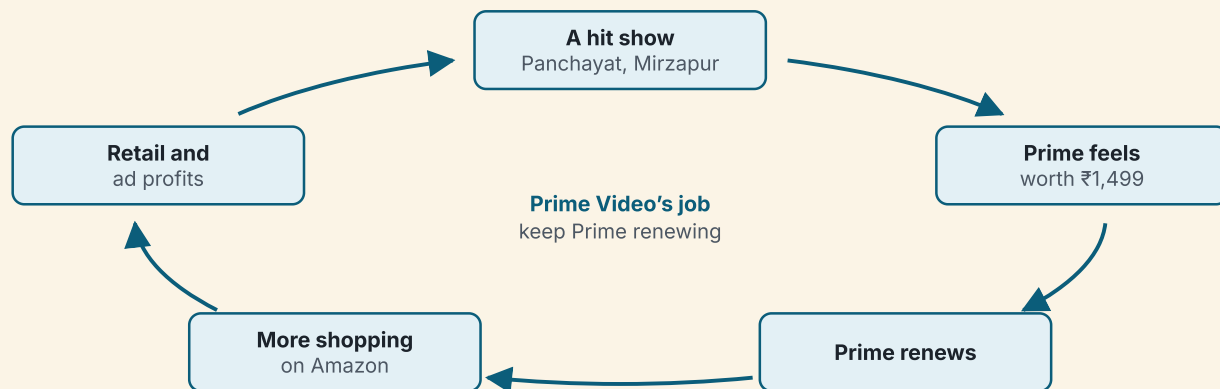


Figure 6.27 · Why Amazon spends on shows it doesn't sell separately. A hit like *Panchayat* pays back through Prime renewals and shopping, not through video revenue. Author's diagram; Prime price and ad-free add-on from Amazon India (2025–26).

STRENGTHS

- ✓ Owns India's biggest live events
- ✓ Every language, every price point
- ✓ Proven at record scale

WEAKNESSES

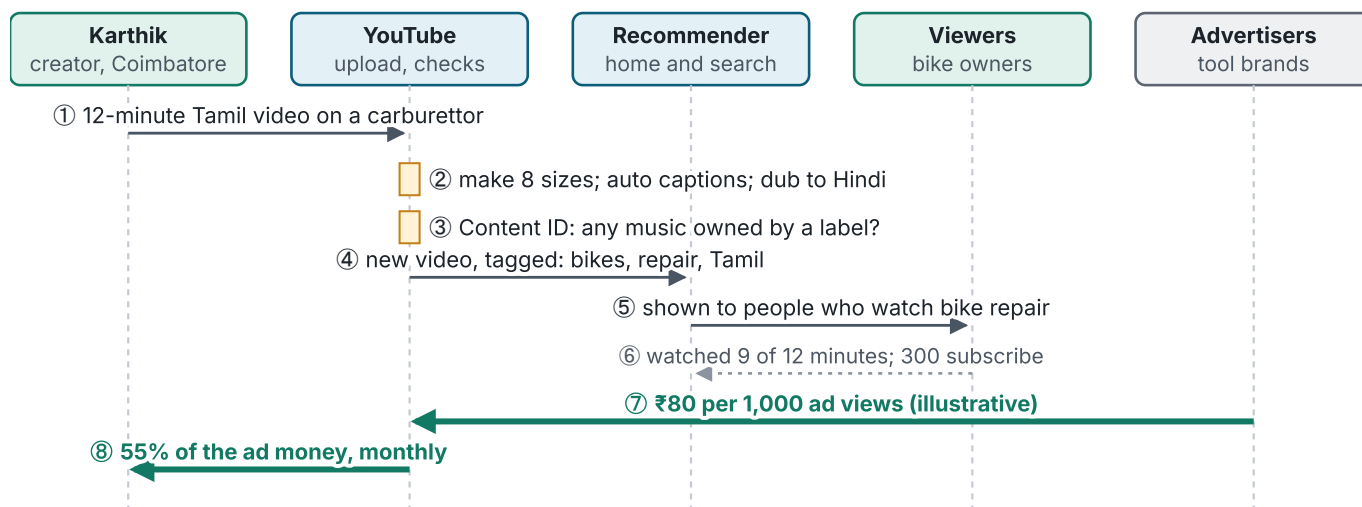
- ✗ Two months of peak a year
- ✗ Rights costs fixed, ad revenue not
- ✗ Many users never chose to pay

What I'd build next

1. Lakhs leave after the IPL → pause instead of cancel, and an “after the IPL” watchlist built during the season → share of IPL joiners watching weekly 60 days after the final.
2. Advertisers pay for reach but want sales → tap an ad during a strategic time-out to save the offer to WhatsApp or JioMart, paid by UPI later → revenue per 1,000 ad views; offer saves per ad.

YouTube: India's free television, made by its viewers

Walkthrough (illustrative persona). Karthik, 34, ran a two-wheeler garage in Coimbatore. He films a 12-minute Tamil video on cleaning a carburettor and uploads it one night. YouTube makes several sizes of it, adds captions and an automatic Hindi dub, and checks the background song against music owned by labels. Then it shows the video to a few thousand people who watch bike-repair videos. They watch nine minutes of twelve on average, so YouTube shows it to more. In a month it has 1 lakh views. Karthik earns about ₹2,640 from the ads on it, more from a tool brand's sponsorship, and more again from channel members who pay him monthly (Figure 6.28).



1 lakh views × 0.6 ads × ₹80 ÷ 1,000 × 55% ≈ ₹2,640 a month from ads (illustrative).

Figure 6.28 · From Karthik's upload to his payout. Content ID (③) protects rights owners before a video spreads; watch time (⑥), not clicks, decides how far it goes. Karthik keeps 55% of the ad money on long videos. Simplified from YouTube's creator help pages (2025–26); India ad price and views illustrative.

Feature	User problem	Why it's designed this way	Metric it moves
Recommendations	"I don't know what to watch"	Ranked on watch time and on surveys that ask "was this worth your time?"	Watch time; satisfaction

Shorts	"I have 30 seconds"	Vertical feed; ad money pooled and shared by views	Daily users; 200 bn Shorts views a day
Partner Programme	Creators: "How do I earn?"	Fan payments from 500 subscribers; ad money from 1,000 subscribers and enough watch hours	Creators earning
Memberships, Super Chat	Creators: "Ads don't pay enough"	Fans pay the creator directly; YouTube keeps 30%	Creator income per channel
Auto-dubbing, captions	"It isn't in my language"	AI dubs a video into other languages	Views from other languages
Premium	"Ads interrupt my music"	No ads, background play, downloads	Paying subscribers

Business model, with numbers. Mostly ads: YouTube's ad revenue was \$36 bn in 2024 and grew 13% in April–June 2026. Then subscriptions (Premium, Music and, in the US, YouTube TV) and a share of fan payments. YouTube says it paid creators, artists and media companies more than \$100 bn from 2021 to 2024. India has about 52 crore YouTube users, more than any other country, but Indian ads are cheap, so it earns far less per Indian viewer. Creators get a bigger share than on most rivals (Figure 6.29).

Creator's share of each rupee (or dollar) earned, 2025–26

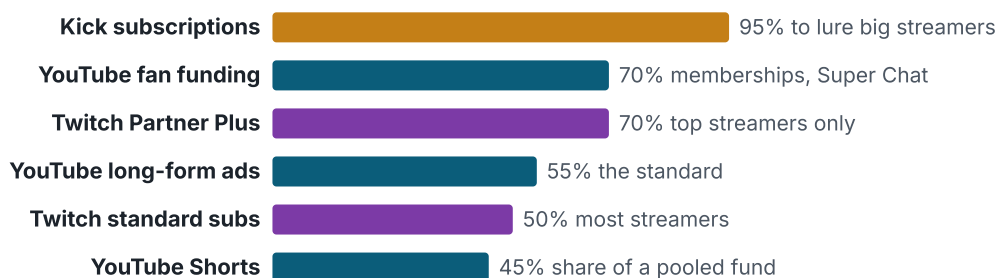


Figure 6.29 · How much of the money goes to the creator. The challenger (Kick) pays the most; the leader pays less but brings the biggest audience. Creators mostly follow the audience. *YouTube, Twitch and Kick creator terms (2025–26); Shorts share applies to a pooled fund.*

Moat, in plain words. Viewers come because creators are there, and creators stay because viewers and money are there. Twenty years of videos is a library nobody can copy. Google's ad sales and Android's reach sit behind it.

NORTH STAR Daily watch time that viewers rate as worth it.	INPUTS Daily viewers · Shorts views · active up-loaders · share of creators earning.	GUARDRAILS Harmful or misleading content · ad load · income concentrated in few creators.
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COMPARISON · MOJ AND TWITCH

Moj (with ShareChat) rose after India banned TikTok in 2020 and has about 15 crore monthly users with ShareChat, across more than 15 Indian languages. It competes with Shorts and Reels by staying local: regional creators, regional ads, and micro-dramas of a few minutes. **Twitch** is the opposite shape: live, long and interactive, with chat as the product. It holds about half of the world's live-streaming watch time, but live video is expensive to deliver; Twitch left South Korea in 2024 over network fees, and rival Kick pays creators 95% to lure them away.

STRENGTHS

- ✓ Largest library and creator base
- ✓ Free, on every phone and TV
- ✓ Pays creators best at scale

WEAKNESSES

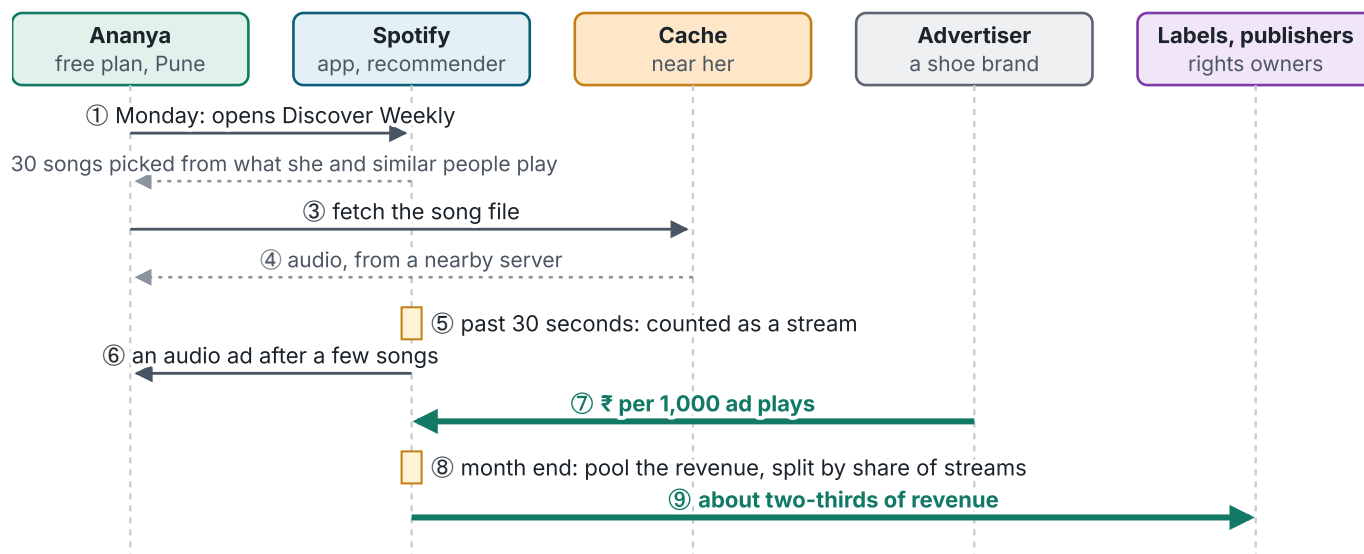
- ✗ Low earnings per Indian viewer
- ✗ Policing harmful content at scale
- ✗ Shorts earn less than long videos

What I'd build next

1. Students watch exam lectures but lose track across channels → turn playlists into courses with progress, notes and short quizzes → weekly learners; share finishing a course.
2. Small creators in Bhojpuri, Odia or Assamese can't reach the earning bar → auto-dubbing plus a fund for fast-growing channels in smaller languages → creators earning, by language.

Spotify: free music that sells Premium

Walkthrough (illustrative persona). Ananya, 20, a college student in Pune, uses Spotify free. Every Monday she opens Discover Weekly, 30 songs picked from what she plays and what people with similar taste play. Most weeks she saves three or four. On the free plan an audio ad plays every few songs, and on her phone she can't always pick the exact song she wants. In December, Wrapped tells her she was in the top 1% of listeners of an indie Hindi band; she shares it on Instagram, and two friends sign up. In January, fed up with ads before her exam playlist, she upgrades (Figure 6.30).



There is no fixed rate per play: each rights owner gets its share of all streams that month.

Figure 6.30 · From a song played to a royalty paid. A play counts as a stream after 30 seconds (⑤). At the end of the month Spotify pools its revenue and splits it by each rights owner's share of all streams (⑧–⑨), rather than paying a fixed rate per play. *Simplified from Spotify's "Loud & Clear" royalty explainer (2024–26); ad price not shown.*

Feature	User problem	Why it's designed this way	Metric it moves
Discover Weekly, Daily Mixes	"I'm bored of my own playlist"	A fresh, personal list every Monday; a small habit	Songs saved; weekly listeners
Wrapped	"What did my year sound like?"	A shareable year in music; free marketing every December	Shares; sign-ups in December
Free tier with limits	"I won't pay yet"	Ads and some limits on choosing songs: good enough to stay, annoying enough to upgrade	Free-to-Premium upgrades
Podcasts, audiobooks	"Something to listen to on the commute"	Costs less per hour than music, which pays per stream	Gross margin
AI DJ, "Talk to Spotify"	"Just play something I'll like"	Voice and chat requests	Listening time

Business model, with numbers. Premium subscriptions bring in most revenue; ads on the free tier bring the rest. In April–June 2026 Spotify had **77.7 crore monthly users and 30 crore paying**, revenue of €4.8 bn and an operating profit of €655 mn. About two-thirds of revenue goes to labels and publishers, which is why its first full year of profit only came in 2024. The free tier is its marketing: most Premium subscribers started free.

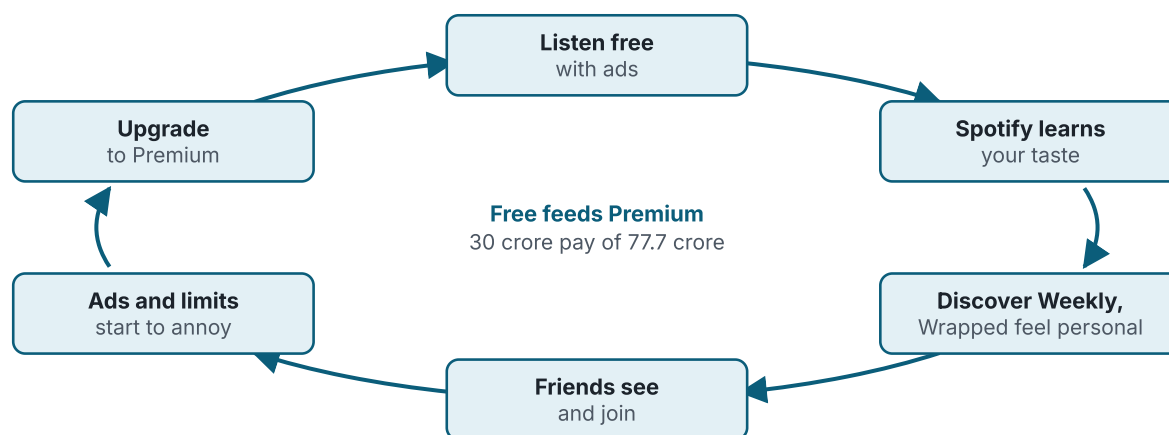


Figure 6.31 · The freemium loop. The free tier teaches Spotify your taste and spreads through Wrapped; the limits and ads then push some listeners to pay. *Author's diagram; user numbers from Spotify Q2 2026 results.*

Moat, in plain words. Years of your playlists and taste live here, which makes switching feel like losing your music. Its size gives it weight with labels. But the labels own the songs, and every rival has the same catalogue, so Spotify competes on picking, not on having.

NORTH STAR Time spent listening by monthly users.	INPUTS Free-to-Premium upgrades · Premium cancellations · podcast minutes · ad slots filled.	GUARDRAILS Gross margin · artists' and labels' satisfaction · ads per hour on free.
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COMPARISON · JIOSAAVN AND THE INDIAN MUSIC APPS

In India, Spotify competes with YouTube Music (free with ads, and bundled with YouTube Premium), JioSaavn and Gaana. JioSaavn is strongest in Hindi and regional film music and reaches people through Jio: its Pro plan comes bundled with some recharges, and the casebook quotes about ₹99 a month for one person. Most Indian listeners never pay; the fight is for free listening hours, which earn from cheap Indian audio ads.

STRENGTHS

- ✓ Best-known personal playlists
- ✓ Huge free funnel into Premium
- ✓ Now steadily profitable

WEAKNESSES

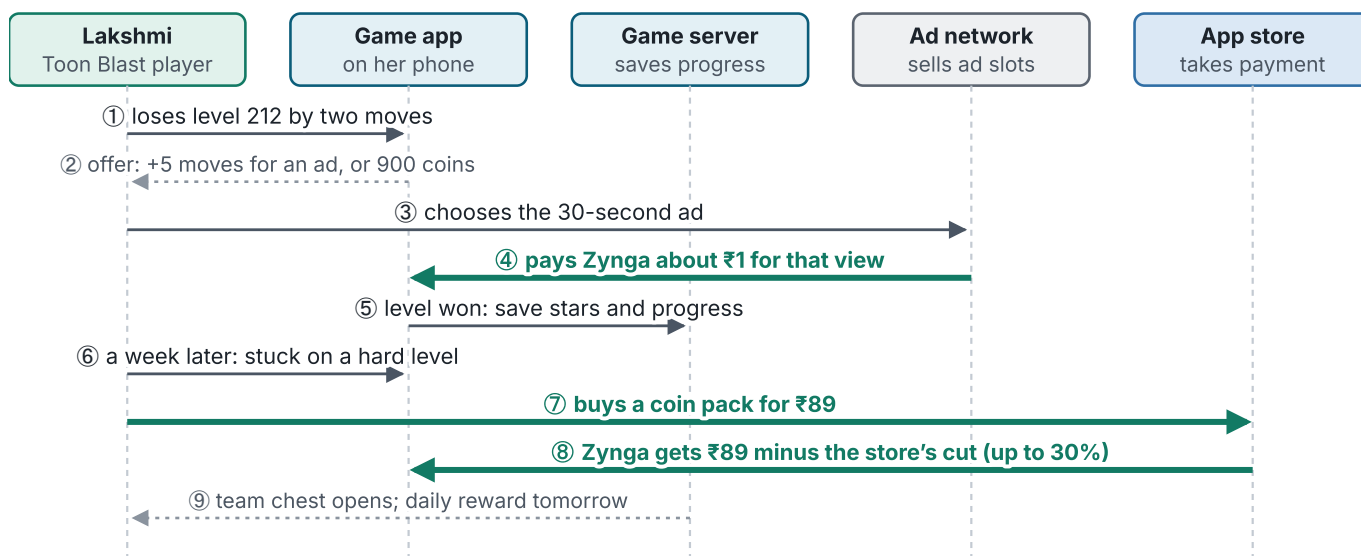
- ✗ Labels take two-thirds
- ✗ Same songs as every rival
- ✗ Few Indians pay for music

What I'd build next

1. Indian free listeners are hard for advertisers to reach by language and city → language radio stations with local audio ads sold to small businesses → free listening hours; India ad revenue per listener.
2. Families share one phone's Premium → a cheaper family plan for two to three people in the same home → paying households; guardrail: individual plans moving down.

Zynga: free to play, paid for by a few

Walkthrough (illustrative persona). Lakshmi, 38, an accountant in Chennai, plays Toon Blast on the train home. She has played for two years and never paid. On level 212 she runs out of moves two cubes short. The game offers five more moves: watch a 30-second ad, or spend coins. She watches the ad and wins. Three seats away, a college student has spent ₹2,400 this month on boosters to beat a level before his friends. The game is the same for both. Its business needs Lakshmi happy and playing, because she makes the teams feel alive and fills ad slots, while the student and players like him pay most of the bills (Figure 6.32).



Most players never pay. Ads earn from them; a few buyers earn most of the rest.

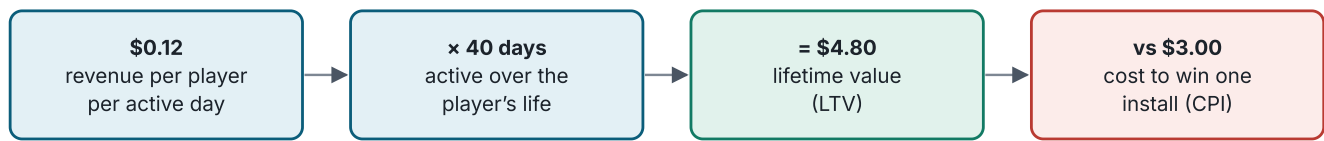
Figure 6.32 · Two ways Lakshmi pays without paying, and one way she might. The “near miss” offer (②) is the moment people are most willing to pay. The app store keeps up to 30% of any purchase (⑧). *Simplified from how casual mobile games generally earn; ad and pack prices illustrative.*

Feature	Player problem	Why it's designed this way	Metric it moves
Short levels, few moves	“A few minutes of progress on the train”	Clear goals, two to three minutes, early levels designed to be won	Level completion; day-1 return
Lives that refill	Game: stop players burning out in one sitting	Limited plays, then a reason to come back later	Sessions per day
Rewarded ads	“I'm stuck but won't pay”	Choose an ad for a bonus; never forced	Ad revenue per player; non-payer retention
Teams, weekly events	“It feels the same every day”	Shared goals, team chests, limited-time events	Day-30 return; spend during events
Web store	Studio: app stores take up to 30%	Sell coins on the web where store rules allow	Margin per purchase

Business model, with numbers. Free-to-play games earn from **in-app purchases** and **ads**. Usually only 2–5% of players ever pay, and a tiny group spends heavily. The test for every game is whether a player is worth more than it costs to win them (Figure 6.33). Zynga began with FarmVille on Facebook in 2009, grew through friends' feeds

and fell when Facebook cut those feeds; it then bought mobile studios with proven hits. Take-Two bought it in 2022 for \$12.7 bn, and in the year to March 2026 mobile was about half of Take-Two's \$6.7 bn of net bookings (money players committed to spend).

Does buying a new player pay back? (illustrative casual-game numbers)



\$4.80 > \$3.00, so ads to win players pay back. Lift day-7 return and the 40 days grows: that is why games obsess over retention.

Figure 6.33 · The one sum a games PM must know. If a player's lifetime value beats the cost of winning them, buying ads to grow pays back. Retention is the lever: more active days raises LTV. *Structure standard in mobile gaming; values illustrative.*

Moat, in plain words. A portfolio of long-running hits, each with years of tuned levels and loyal teams; and the data to know which players will pay and which ads win players cheaply. Lesson from FarmVille: growth that depends on someone else's platform can be switched off by that platform.

NORTH STAR Daily active players who come back the next week.	INPUTS Day-1, day-7, day-30 return · share who ever pay · revenue per daily player · cost per install.	GUARDRAILS Spending by very heavy payers (fair treatment) · difficulty complaints · ad frequency.
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WHEN IT BREAKS · INDIA'S BAN ON REAL-MONEY GAMES
The Online Gaming Act 2025 banned games played for money from 1 October 2025, including fantasy cricket and rummy, and banned their ads and payments. Casual games like Toon Blast, where you buy extras but can't win money, were not affected. The line that matters is simple: can a player put money in *and take money out*? Any game design that looks like a bet now carries legal risk in India.

STRENGTHS

- ✓ Many long-lived hits
- ✓ Skilled at buying players profitably
- ✓ Take-Two's money behind it

WEAKNESSES

- ✗ Hits are rare and fade
- ✗ Depends on app stores and ad networks
- ✗ Heavy spenders invite scrutiny

What I'd build next

1. Indian players rarely pay ₹89 at once → ₹10 “sachet” boosters paid by UPI → share of players who ever pay; revenue per daily player.
2. Players leave when stuck on a hard level for days → spot players stuck three days and quietly give a free booster → day-30 return; guardrail: purchases per payer.

Side by side

	Netflix	JioHotstar	YouTube	Spotify	Zynga
Main job	Something good to-night, found fast	The match live, in my language	Entertain or teach me, free	Music that fits my mood	A few minutes of progress
Who pays	Viewers, monthly	Advertisers; viewers, often via recharge	Advertisers; some viewers	Premium listeners; advertisers	A few players; advertisers
Biggest cost	Shows, paid once	Sports rights	Creators' share	Royalties per stream	Buying new players
Latest number	\$12.6 bn revenue, Q2 2026	53 crore monthly users	~52 crore Indian users	30 crore paying	Mobile ≈ half of \$6.7 bn
Moat	Content scale, data	Cricket rights, Jio	Creators and viewers	Your playlists	Portfolio of hits
Biggest threat	No big live sport in India	Rights cost vs ad market	Harmful content rules	Labels' power	Store fees; fading hits

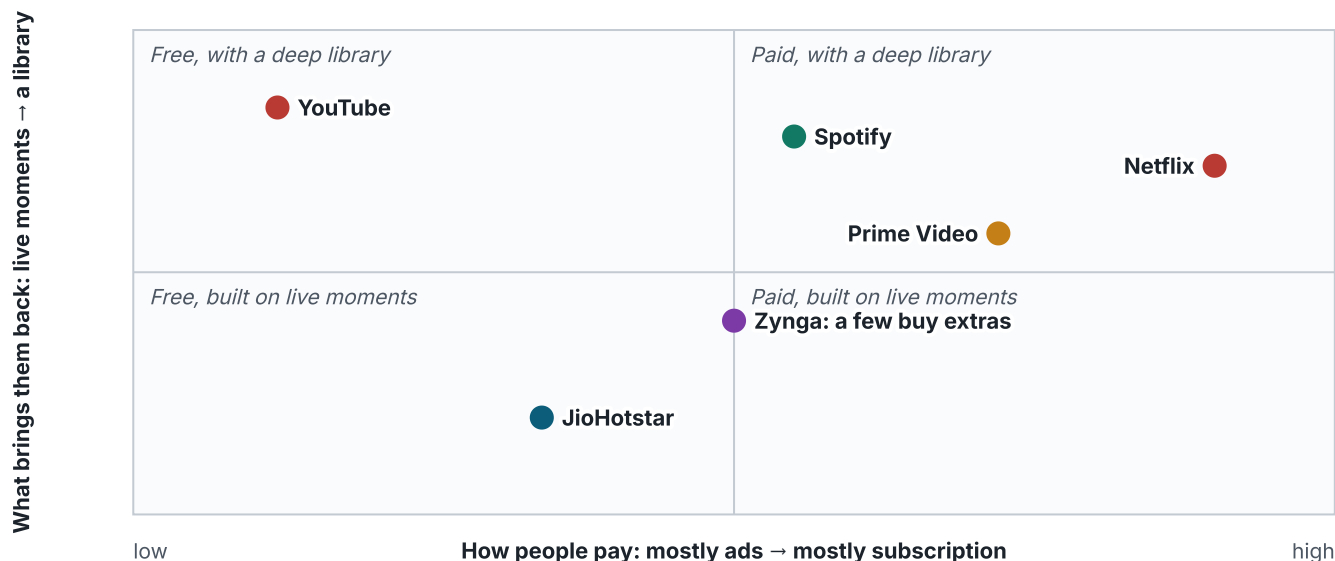


Figure 6.34 · How people pay, and what brings them back. Live events pull people in on a date; a library keeps them between events. Paid services need a library, because people won't pay every month for two months of cricket. *Author's placement; positions are judgement, not measured values.*

Five ways to earn, and the one number each must get right

Model	Example	The number that decides it	What goes wrong
Subscription	Netflix	Monthly churn: 3% loses a third of members a year; 6% loses over half	People join for one show and leave
Ads	YouTube, JioHotstar	Minutes × ads per minute × price per 1,000	Too many ads push viewers away; a weak ad year
Bundle	Prime Video, Jio recharges	Whether the partner's customers renew or buy more	Users who never chose you, and never watch
Freemium	Spotify	Share of free users who upgrade each year	Free too good, or too annoying
In-app purchases	Zynga	Lifetime value against the cost of winning a player	Hits fade; heavy spenders draw regulators

IN THE INTERVIEW · HOW THESE PRODUCTS COME UP

- **“Why doesn't Netflix show IPL?”** Rights cost about ₹58 crore a match for digital alone; that only pays back with a mass ad business, which Netflix doesn't run in India.
- **“How does Spotify make money if most users are free?”** Free listeners see ads and many later upgrade; Premium brings most revenue; labels take about two-thirds.
- **“Why does Prime Video exist?”** To make Prime renew and keep people shopping on Amazon; its ads now use shopping data.
- **“Design a monetisation feature for a free game.”** Rewarded ads for non-payers, small packs at the near-miss moment, and a guardrail on heavy spenders.

Blockchain: a shared record that nobody can quietly change

Twenty families in a Pune housing society run a savings group, a *bhishi*. Each month every family puts in ₹5,000 and one family takes the ₹1 lakh pot. Mrs Kulkarni, the secretary, keeps the account book. In the third year, a family says it paid in March; the book says it didn't. Did it forget, or did someone change the page? Nobody can prove either. Now imagine that every family keeps its own copy of the book, every entry is read out and written in all twenty copies, and each page carries a seal made from the page before it. To cheat, you would have to rewrite the same page in more than half the books, and every page after it, without anyone noticing. That is the whole idea of a blockchain. Making it work takes a seal that shows any change, a fair rule for who writes the next page, and a reason good enough to justify the effort.

One savings group, eight ideas

Idea	The society's savings book	A blockchain
Ledger	The account book of who paid whom	A list of every transaction ever made
Node	Each family's copy of the book	Each computer that keeps a full copy
Block	One page of entries	A batch of transactions, written together
Hash, chain	A seal on each page, made from the page and the previous seal	A fingerprint of each block, stored in the next block
Consensus	The rule for who writes the next page, and how the others check it	Proof of work (solve a puzzle) or proof of stake (put down a deposit)
Wallet, keys	Your signature, which only you can make	A private key that signs your payments; an address anyone can pay into
Smart contract	A rule in the book that carries itself out: "the pot goes to whoever's name is drawn"	A small program on the chain that moves money when conditions are met
Fee	Tea for the person who writes the page	A "gas" or transaction fee paid to whoever adds the block

1. A block: one page of entries

Everyday picture. One page of the society book: the date, three entries, and the seal from the previous page at the top.

How it works. A block holds a batch of transactions, a time stamp, the fingerprint of the previous block, and its own fingerprint (Figure 6.35). The fingerprint is a **hash**: a short code worked out from the block's contents, the same kind of fingerprint apps use to store passwords safely. The same contents always give the same hash, and changing even one character gives a completely different one. Real SHA-256 outputs: "Hello" gives 185f8db3..., "hello" gives 2cf24dba.... One capital letter, a new fingerprint.

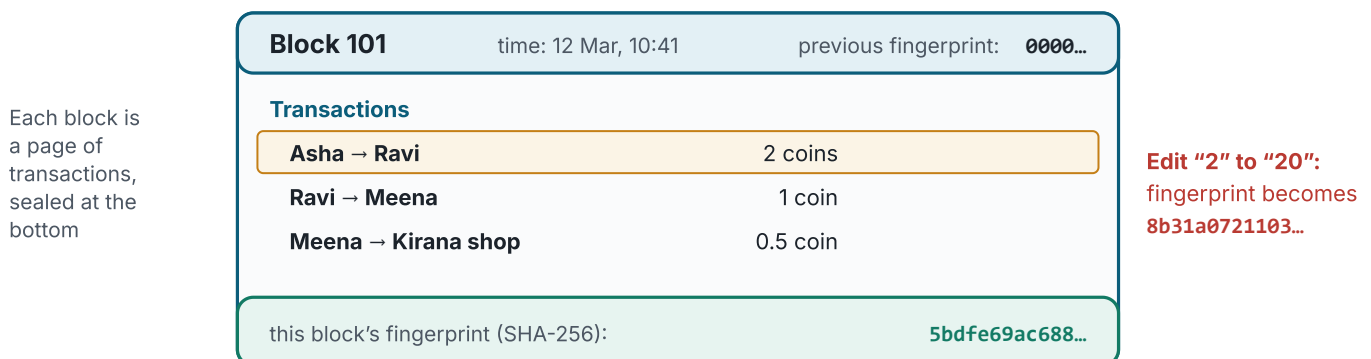


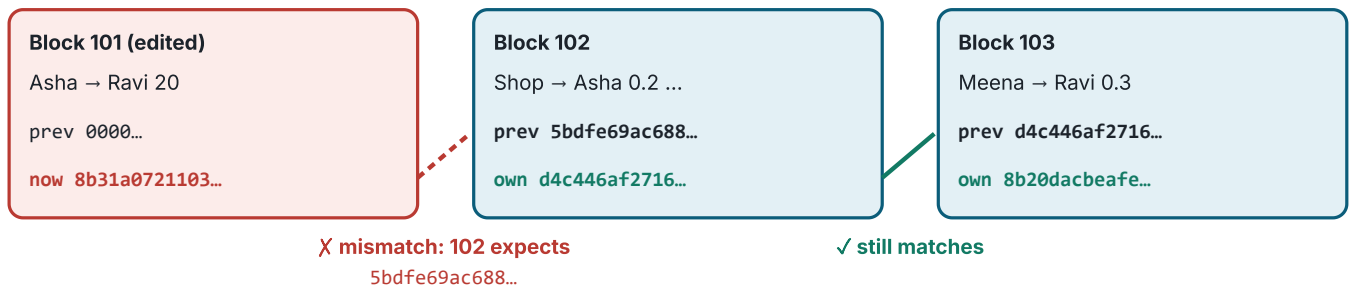
Figure 6.35 · One block, three payments. The block stores the previous block's fingerprint (top) and has its own (bottom). Change Asha's 2 coins to 20 and the fingerprint changes from 5bdf69a... to 8b31a072.... Real SHA-256 fingerprints of the text shown, first 12 characters; names and amounts illustrative.

2. The chain: why old pages can't be changed quietly

Everyday picture. Each page's seal is pressed partly over the previous page's seal. Tear out page 12 and rewrite it, and the seal on page 13 no longer lines up.

How it works. Block 102 contains block 101's fingerprint. If someone edits block 101, its fingerprint changes, so block 102 now points at a fingerprint that doesn't exist (Figure 6.36). To hide the edit, the cheat must redo 102, then 103, and every block after it, and do it on more than half of all the copies, faster than honest writers add new

blocks. On a large network, that is practically impossible. This property is called being **tamper-evident**: changes are not impossible, but they are always visible.



To hide the edit, the cheat must redo 102, 103 and every later block, on over half of all copies.

Figure 6.36 • Change one old entry and every later link breaks. Block 102 still says the previous fingerprint is 5bdf69a..., but the edited block 101 now hashes to 8b31a072.... Every honest copy spots the mismatch. *Real SHA-256 fingerprints (first 12 characters) of the illustrative blocks.*

Real example. Bitcoin's ledger has run since January 2009 with no one able to rewrite an old block. **PM decision:** tamper-evidence is valuable only when people fear the record keeper might change the record. If everyone trusts the keeper (a bank, the RBI, the railways), a normal database with an audit log gives the same comfort for a tiny fraction of the cost.

3. Consensus: who gets to write the next page?

Everyday picture. If any family could write the next page whenever it liked, two families could write different pages at once, or one could write only entries that suit it. The society needs a fair rule for choosing the writer, and a way for everyone to check the page.

How it works, proof of work. Bitcoin runs a lottery that costs electricity to enter. To write the next block, a computer must find a number (a **nonce**) that, added to the block, makes its hash start with a set number of zeros. There is no shortcut: you guess, hash, and check. On a laptop, making block 102's hash start with four zeros took **31,312 guesses** (the average is $16^4 = 65,536$). Anyone can check the winner's answer with one hash. Bitcoin sets the target so the whole network finds one block about every 10 minutes; in 2025 that meant roughly 10^{23} guesses per block (Figure 6.37). Cheating means out-guessing everyone else combined, which needs more than half the world's mining power: a **51% attack**.

How it works, proof of stake. Instead of burning electricity, writers lock up a deposit of coins. The network picks the next writer at random, weighted by deposit, and others check the block. A writer who signs two different blocks or lies loses part of the deposit (**slashing**). Ethereum switched from work to stake on 15 September 2022 and cut its electricity use by more than 99.9%.

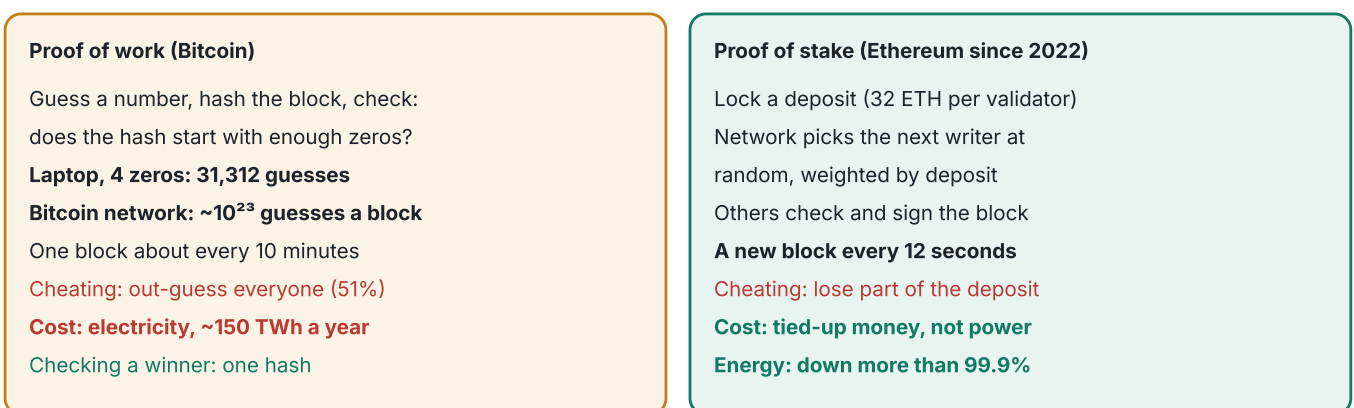


Figure 6.37 • Two ways to choose the writer. Proof of work makes cheating expensive in electricity; proof of stake makes it expensive in lost deposits. Both let any stranger check the result. *Bitcoin network difficulty (2025, order of magnitude); Ethereum Foundation (the Merge, 15 September 2022); laptop test by the author.*

Real example, and the cost. Cambridge University estimates Bitcoin uses around 150 terawatt-hours a year, about as much as a mid-sized country, for roughly 4–5 lakh transactions a day. That works out to roughly 900 kilowatt-hours per transaction, close to what an average Indian home uses in a year (estimate). A UPI payment uses a tiny fraction of one kilowatt-hour. Small chains have suffered real 51% attacks: Ethereum Classic had blocks rewritten several times in 2020. **PM decision:** the consensus method decides cost, speed and energy. Bitcoin handles about

7 payments a second; UPI handles thousands. If a product needs speed and low fees for everyday payments, a public proof-of-work chain is the wrong base.

4. Wallets and keys: “not your keys, not your coins”

Everyday picture. A letterbox outside your flat. Anyone who knows the address can drop a letter in; only the person with the key can open it and take things out.

How it works. A crypto wallet holds a **private key**, and from it an **address** that others can pay into, a public/private key pair (two linked keys: what the private key signs, anyone can check with the public one). To spend, the wallet signs the payment with the private key; every node checks the signature with the public key. There is no password reset: the key is usually backed up as a **seed phrase** of 12 or 24 words. Lose it, and the coins are gone for ever. Share it, and they can be taken for ever. Chainalysis has estimated that around a fifth of all bitcoins are lost or stuck in wallets nobody can open. Many people let an exchange hold their keys instead (Figure 6.38). That is easier, but it means trusting the exchange: in July 2024 hackers took about \$230 million from the Indian exchange WazirX, and users’ withdrawals were frozen for months.

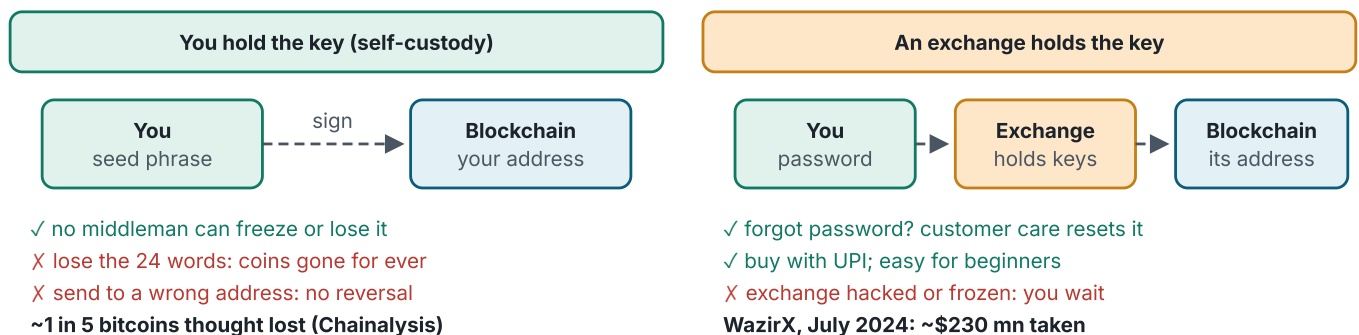


Figure 6.38 · Who holds the key decides who you must trust. Holding your own key removes the middleman but leaves you alone with mistakes; an exchange brings back customer care and brings back the risk of a trusted party. *Author’s diagram; WazirX incident (July 2024, widely reported); Chainalysis estimate of lost bitcoin.*

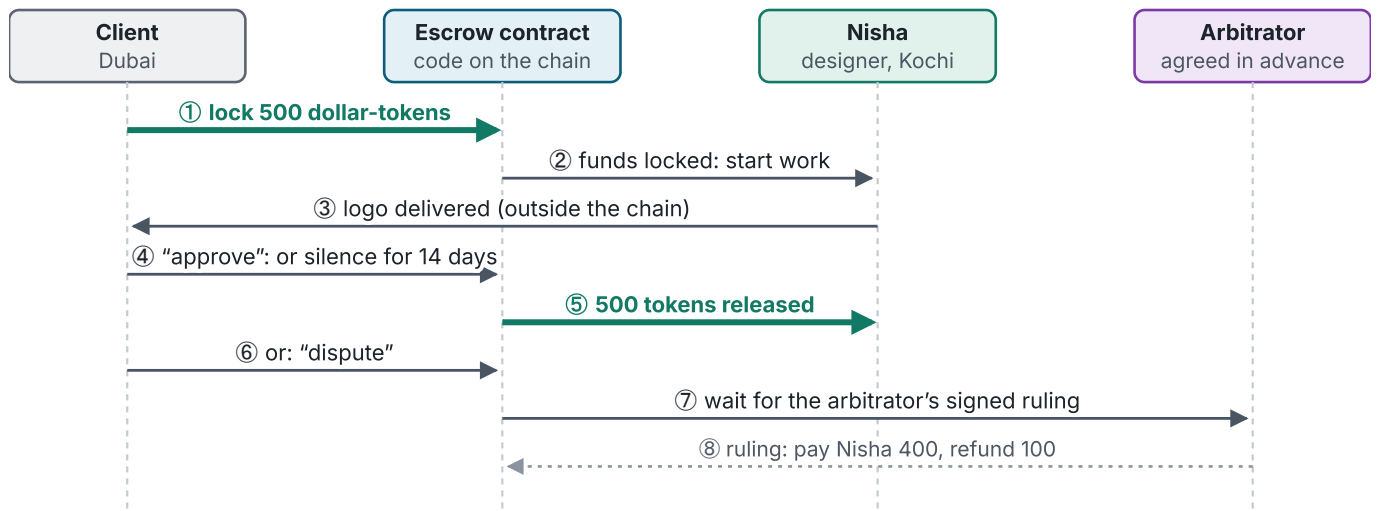
PM decision: key recovery is the biggest user-experience problem in crypto. Every design choice (a seed phrase, recovery through friends, an exchange holding the key) trades freedom from middlemen against the chance of losing everything.

5. Smart contracts: a vending machine made of code

Everyday picture. A vending machine at a railway station. Put in ₹20, press B4, and a bottle drops out. No shop-keeper decides; the rules are built into the machine.

How it works. A **smart contract** is a small program stored on a blockchain (usually Ethereum) that holds money and moves it when conditions are met. Worked example, an escrow (Figure 6.39): Nisha, a designer in Kochi, agrees to make a logo for a client in Dubai for 500 dollar-tokens (a **stablecoin**: a token meant to stay worth one dollar). The client locks 500 tokens in the contract. When Nisha delivers and the client clicks “approve”, the contract pays her. If the client goes silent for 14 days after delivery, it pays her anyway. If the client raises a dispute, it waits for an agreed arbitrator. Nobody can run off with the money in between.

Where it breaks. Code does exactly what it says, bugs included, and on a blockchain you can’t quietly patch it. In June 2016 an attacker used a flaw in “The DAO”, a smart-contract investment fund, to drain about 3.6 million ether, then worth around \$50 million. Ethereum’s community reversed it only by splitting the whole chain in two. And a contract only knows what is on the chain: a delivery, a match result or a price must be fed in from outside (the **oracle problem**), and whoever feeds it in must be trusted. **PM decision:** smart contracts need audits before launch, limits on how much money they hold, and a plan for disputes, because “code is law” also means “bugs are law”.



The contract holds the money fairly, but it can't judge the logo: that still needs a person.

Figure 6.39 • An escrow contract. The money sits in code, not with either person. But the contract can't see the logo: whether the work was good still needs a human or an outside source of truth. *Illustrative example.*

6. Real uses and real failures

Use	What happened	The lesson
Crypto payments	Bitcoin was built as money without banks. In India, gains on crypto are taxed at 30% with 1% tax deducted on every transfer (since 2022); prices swing widely. Stablecoins are used more for moving dollars abroad than for shopping	Works where people distrust the currency or the banks; UPI is faster and cheaper for everyday Indian payments
Food tracing	Walmart and IBM traced a packet of sliced mangoes to its farm in 2.2 seconds, against about 7 days on paper (2018 pilot)	Worked because Walmart could require its suppliers to join. The speed came from digital records; the chain added shared trust
Shipping (TradeLens)	Maersk and IBM built a shared shipping ledger in 2018 and shut it in early 2023: rival shipping lines wouldn't join a platform run by a competitor	A shared ledger fails if one rival controls it; the hard part is getting rivals to agree, not the software
e-rupee (RBI)	RBI pilots since late 2022: about 80 lakh users; ₹772 crore in circulation in March 2026, down 24% in a year; programmable food subsidies in Gujarat, Puducherry and Chandigarh	A digital rupee needs no public blockchain: the RBI is the trusted party. Hardly anyone switched from UPI
NFTs	Digital art tokens boomed in 2021 (a Beeple work sold for \$69 million at Christie's) and trading collapsed by more than 90% within two years	The token proves who owns a record, not a right to the art or any lasting value
Music royalties	Start-ups promised artists paid per play on a chain. Few labels joined, and play counts still came from the streaming apps	The oracle problem again: the chain is only as honest as the data fed into it

7. Public or private: who may join, and who may write

Everyday picture. A notice board in a railway station, which anyone may read and anyone may pin to, against a housing society's minutes book, which only members see and only the committee writes in.

How it works. Two questions sort every blockchain. Who may **read and join**: anyone, or only approved members? And who may **write and check** new blocks: anyone, or only approved members? A **public, permissionless** chain (Bitcoin, Ethereum) lets anyone do both; that is what makes it hard to censor, and also what makes it slow and costly. A **private, permissioned** chain (built on tools such as Hyperledger Fabric or R3 Corda) lets only named organisations join and write. It is fast and keeps data private, but it only works if members trust the group that approves them.

	Anyone may write and check	Only approved members write and check
Anyone may read	Public, permissionless: Bitcoin, Ethereum. Open, slow, costly per transaction	Public, permissioned: a registry anyone can inspect but only approved bodies update
Only members read	Private, permissionless: rare in practice	Private, permissioned: banks sharing trade papers; a consortium of exporters. Fast, private

PM decision: a private chain answers most business needs, but ask the awkward question: if a handful of known members run it, why not one shared database with signed entries and an audit log? The honest answer must be "because no member will let another run it", as TradeLens found out.

8. When a blockchain is not the answer

A blockchain is a slow, costly database with one special power: it lets people who don't trust each other, and don't trust any one keeper, share a record nobody can quietly change. Five questions test whether you need that power (Figure 6.40). Most proposals fail at the second or third.

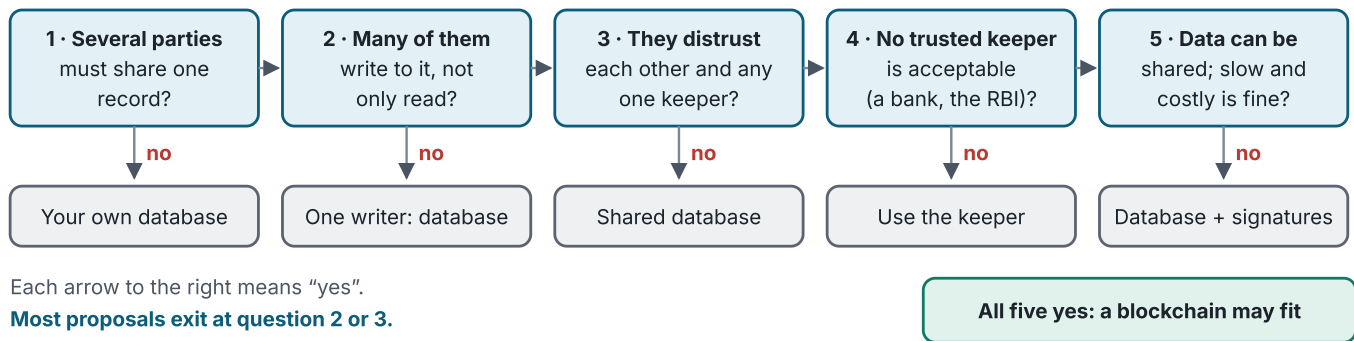


Figure 6.40 · Do you need a blockchain? Answer in order. If any answer sends you left, a normal database (perhaps with an audit log and digital signatures) does the job faster and cheaper. *Adapted from the World Economic Forum's blockchain toolkit and NIST's blockchain overview (2018–20).*

Proposal	Many writers?	No trusted keeper?	Can be public, slow?	Verdict
IPL tickets on a chain to stop touts	✗ one seller	✗ the ticketing app	✓	Database with names on tickets and ID at the gate
A state's land records	✗ one registrar	✗ the state	✗ private data	Database with signed entries and a public audit log
College degree certificates	✗ each college	✗ each college	✓	Digitally signed certificates, as DigiLocker already does
Trade papers between banks in six countries	✓	✓ no bank trusts one bank	✓ within the group	A private, permissioned chain can fit
Royalties for a streaming app's songs	✗ the app counts plays	✗	✓	Publish audited play counts instead

IN THE INTERVIEW · HOW BLOCKCHAIN COMES UP

- **"Should we use blockchain for X?"** Ask who writes, who they don't trust, and whether one keeper would do. Name the oracle problem: the chain can't check that the data was true when entered.
- **"Explain blockchain to a ten-year-old."** A class notebook that every student copies, where each page is sealed with the one before, so nobody can change an old page without everyone noticing.
- **"Why hasn't the e-rupee taken off?"** UPI already moves money free and instantly; the e-rupee must offer something UPI can't, such as money programmed for one purpose.

System design V: one IPL final, to crores of phones at once

7:28 pm, final night. In the next five minutes, several crore people will open JioHotstar, log in, perhaps renew a plan, and ask for the same few seconds of video at the same moment. Priya's phone on the hostel steps has a patchy 4G signal; her father's smart TV in Rajkot is on fibre; a truck driver near Nagpur is on a ₹199 data pack. Each of them should get the best picture their connection can carry, the match should never stop, and the ads in every over-break must be sold and shown. Follow one ball from the stadium to those three screens.

One dairy, ten ideas

Getting live video to crores of people works much like getting milk to a city every morning. Each row of this table is one of the ideas that follow.

Idea	The dairy	Live video
Concurrency	Everyone wants milk between 6 and 7 am	Crores want the same ball at the same second
Bitrate ladder	The same milk in 200 ml, 500 ml and 1-litre packs	The same match at 240p to 4K
Segments	Milk sent in crates, not one giant tank	Video cut into chunks of a few seconds
Adaptive bitrate	The milkman brings a smaller pack when the lift is broken	The player picks a lower quality when the network slows
CDN, edge cache	Chilling centres and neighbourhood booths near homes	Servers in every city, even inside internet providers
Latency	Time from cow to doorstep	Seconds from the stadium camera to your screen
Pre-warming	Extra tankers booked before Diwali, not on the day	Servers switched on before the toss, not when the rush hits
Ad insertion	A leaflet slipped into each packet, different for each street	Different ads stitched into each viewer's stream
Graceful degradation	In a shortage, stop selling cream and ghee; never stop milk	Drop 4K and extras first; never stop the match
DRM, watermarks	A sealed pack with a batch number, to trace adulteration	Encrypted video with a hidden viewer mark, to trace pirates

The load, in numbers

Suppose 6.5 crore phones and TVs watch at once, the March 2026 record. At an average of 1.5 megabits per second each, that is $6,50,00,000 \times 1.5 \text{ Mbps} \approx \mathbf{100 \text{ terabits per second}}$, more than many countries' entire internet traffic, all carrying the same few seconds of video. Live is harder than on-demand in three ways. Everyone wants the *same* moment, so nothing can be sent the night before. The crowd arrives in minutes, not over an evening. And a failure is public and remembered: the wicket everyone missed.

1. Encoding and the bitrate ladder

Everyday picture. A dairy sells the same milk in 200 ml pouches for a single person and 1-litre packs for a family. Nobody gets different milk; they get the size that suits them.

How it works. The stadium feed arrives at the platform as one very high-quality stream. **Encoders** (machines that compress video) turn it into several versions at once, the **bitrate ladder**: each rung is a picture size and a number of bits per second (Figure 6.41). The data a viewer uses is simple arithmetic: megabits per second $\times 3,600 \text{ seconds} \div 8 = \text{megabytes an hour}$. At 1.5 Mbps, $1.5 \times 3,600 \div 8 = \mathbf{675 \text{ MB an hour}}$; a three-and-a-half-hour match is about 2.4 GB, most of a ₹199 pack's daily data. At 0.4 Mbps it is 180 MB an hour. Newer **codecs** (the compression methods, such as HEVC and AV1) squeeze the same picture into 30–50% fewer bits, but older phones can't play them, so platforms encode in more than one.

rung	megabits per second	data an hour	who gets it
240p	0.4	180 MB	weak 4G, data saver
360p	0.8	360 MB	most phones on 4G
480p	1.2	540 MB	phone, good signal
720p	2.5	1,125 MB	phone on Wi-Fi, small TV
1080p	5	2,250 MB	TV on broadband
4K	15	6,750 MB	big TV on fibre

MB an hour = Mbps × 3,600 ÷ 8. A 3½-hour match at 1.5 Mbps ≈ 2.4 GB.

Figure 6.41 • One match, many sizes. Each rung is the same picture at a different quality. Most Indian phone viewers sit on the middle rungs; 4K is for TVs on fibre. *Typical ladder values used across the industry (illustrative); data arithmetic exact.*

Real example. JioCinema and then JioHotstar showed IPL in 4K on TVs while keeping low rungs for phones on weak signals; data-saver options let viewers cap quality to save their packs. **PM decision:** which rungs to offer and what the default is. A higher default looks better but eats viewers' data and the company's delivery bill; a lower default is cheap but looks poor on TVs. Many apps pick by device: low on phones, high on TVs.

2. Segments and adaptive bitrate

Everyday picture. Milk goes out in crates, not in one giant tanker to each house. And a good milkman, seeing the lift broken, brings two small packs instead of one heavy can.

How it works. Each rung is cut into **segments** of 2–6 seconds. The player first downloads a small list (the **manifest**) of all rungs and segments, then fetches one segment at a time, using standard formats called HLS and DASH. After each segment, it checks how fast it arrived and how much video is waiting in its **buffer**. If a 4-second segment at 1.5 Mbps arrived in 2 seconds, the network could carry about 3 Mbps, so the player steps up. If segments start arriving slower than they play, it steps down before the buffer empties. This is **adaptive bitrate** (ABR) (Figure 6.42). The buffer is the safety margin: 12 seconds of stored video gives the player 12 seconds to react when Priya's train enters a tunnel.

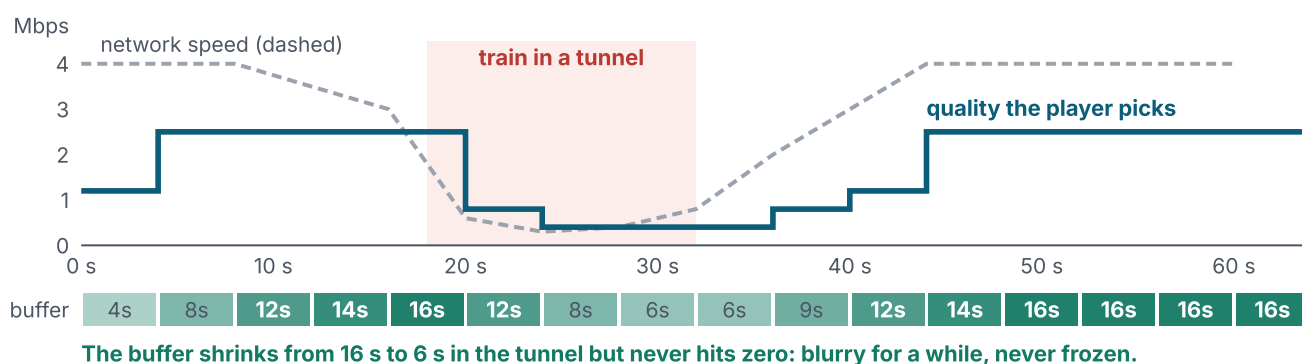


Figure 6.42 • A train journey, as the player sees it. When the network dips, the player drops quality and lives on its buffer; the picture blurs for a few seconds but never freezes. Freezing is what viewers punish. *Illustrative trace.*

Real example. Research on streaming (Conviva and university studies, 2011–13) found that a 1% increase in time spent frozen cut viewing of a live event by several minutes. That is why players prefer a blurry picture to a pause. **PM decision:** how cautious the player is. A cautious player (big buffer, slow to step up) rarely freezes but looks soft; an aggressive one looks sharp but freezes more. For live sport, the usual choice is caution.

3. CDNs and edge caching

Everyday picture. A dairy doesn't send a truck from the plant to every home. Milk goes to chilling centres, then to neighbourhood booths; each booth serves a few hundred homes. If the booth runs out, it fetches more from the chilling centre, not from the plant.

How it works. A **CDN** (content delivery network) is thousands of servers placed near viewers, in every big city and often inside the internet providers' own buildings. The first viewer in Nagpur to ask for a segment makes the Nagpur server fetch it from further up; every other viewer in Nagpur gets the copy it already holds. The share of requests answered from the nearby copy is the **hit rate**, and it decides how much reaches the source (the **origin**) (Figure 6.43). With 6.5 crore viewers each asking for a segment every 4 seconds, about 1.6 crore requests arrive a second. At a 95% hit rate, 5% reach the next layer: 8 lakh a second. At 99%, only 1.6 lakh. Four points of hit rate,

five times less load. Platforms add an **origin shield**, a middle layer that collects the misses so the origin sees each segment asked for only a few times.

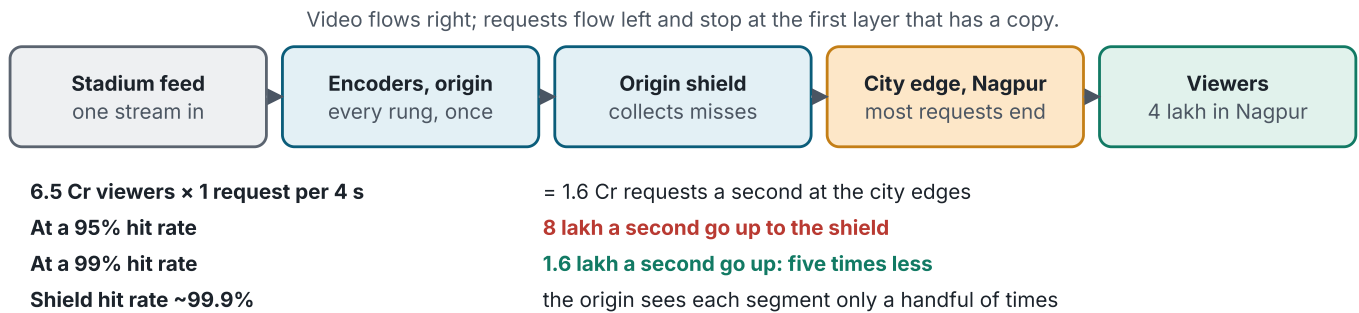


Figure 6.43 · From the stadium to a street in Nagpur. Each layer answers most requests itself; only misses travel up. Moving from a 95% to a 99% hit rate cuts the load on the layer above five times. *Illustrative numbers; the structure is standard for large streaming services.*

Real example. Netflix places its own cache servers (Open Connect) inside internet providers, including Indian ones, and fills them overnight with the shows people are likely to watch the next day. Large live platforms use several CDN companies at once and move traffic between them if one struggles. **PM decision:** how much to spend on delivery, and where. Caches inside providers cost money and negotiation but cut delays and bills; using two CDNs costs more but removes a single point of failure on final night.

4. Live latency: why your father sees the wicket first

Everyday picture. Milk from a village reaches a city home in hours: collection, chilling, packing, transport, the booth, the doorstep. Each step is short; together they add up.

How it works. **Glass-to-glass latency** is the delay from the camera lens to your screen (Figure 6.44). Production and graphics take a few seconds; encoding a few more; the player waits for whole segments to be ready; the CDN adds a little; and the player keeps a buffer of two or three segments for safety. With 6-second segments, that adds up to about 30 seconds. Satellite TV is usually 5–10 seconds behind the stadium. That gap is why Priya's father phones about a wicket she hasn't seen. **Low-latency** streaming cuts segments into parts of about a second and shrinks the buffer, reaching 5–8 seconds, but a smaller buffer means more freezes on weak networks.

App with 6-second segments: about 29 seconds, glass to glass



Satellite TV: about 7 seconds



Low-latency mode (1-second parts, small buffer): about 6 seconds, but more freezes on weak networks



Figure 6.44 · Where the 30 seconds go. Most of the delay is waiting for segments and the player's safety buffer. Shrinking them cuts delay but raises the risk of freezes. *Illustrative budget for 6-second segments; typical satellite delay.*

PM decision: delay against smoothness. For a match with neighbours cheering and WhatsApp buzzing, spoilers hurt; for a film, delay doesn't matter at all. Many platforms offer lower delay on good connections and keep a bigger buffer on weak ones. Anything interactive, such as live predictions or voting, must allow for viewers being 30 seconds apart.

5. The toss-time spike

Everyday picture. Before Diwali a dairy books extra tankers and staff a week ahead. If it waited until the morning queues formed, the extra tankers would arrive after the customers had gone.

How it works. Viewers don't arrive smoothly: millions join in the five minutes before the toss, and again when a star walks out to bat (Figure 6.45). Each new viewer logs in, loads the home page, perhaps pays for a plan, then starts the video, so the first wave hits login, payments and the home page, not the video. **Autoscaling** (adding servers automatically as load rises) takes minutes, too slow for a surge that arrives in seconds. So the platform **pre-warms**: it starts servers before the match for the crowd it expects, from past matches and the teams playing. It also serves the home page from caches, lets a phone that is already logged in skip login, and, if payments pile up, lets viewers start watching while the payment finishes.

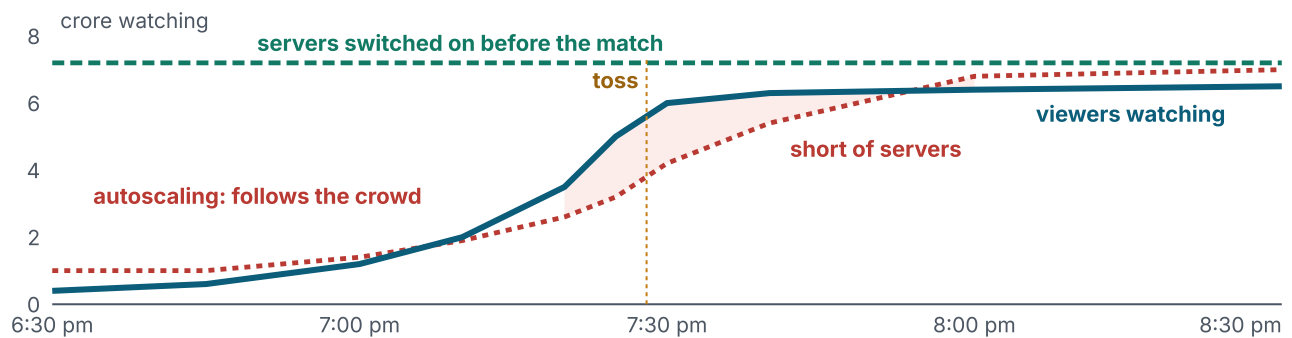


Figure 6.45 · The crowd arrives faster than servers can start. Capacity added by autoscaling (red) lags the surge; capacity switched on before the toss (green) is ready when viewers arrive. *Illustrative curve; the approach follows Hotstar engineers' public talks on IPL scaling (2019–20).*

Real example. Hotstar's engineers described in public talks how they scaled ahead of big matches using forecasts, because waiting for autoscaling left them short at the toss, and how they had switches to turn off heavy features when load passed set levels. **PM decision:** how much spare capacity to pay for. Over-provision and you pay for idle servers on a dull match; under-provision and the final breaks in front of crores. The forecast (which teams, which stage, which time) belongs to the product team as much as to engineering.

6. Ads inside a live match

Everyday picture. A dairy that slips a local shop's leaflet into each milk packet, a different leaflet for each street, all in the half-hour before dawn.

How it works. At the end of an over, the production team sends a marker saying "break starts now, 30 seconds". Each viewer's stream is then filled with ads chosen for that viewer, stitched in on the server so the ads look like the match to the phone: **server-side ad insertion**. The hard part is the timing. If 6.5 crore players all ask "which ads for me?" in the same second, the ad system faces 6.5 crore decisions at once. So platforms spread the questions over the seconds before the break, and prepare likely ads in advance.

PM decision: how personal to make the ads against how reliable the break must be. A fully personal choice for every viewer earns more per ad; a few broad groups (city, language, device) are far easier to serve at peak. If the ad system can't answer in time, the fallback must be a pre-chosen ad or the match, never a black screen.

7. Graceful degradation: what to switch off first

Everyday picture. In a milk shortage, the dairy stops making cream, curd and ghee before it cuts a single family's daily milk.

How it works. The team agrees a ladder in advance (Figure 6.46): as load rises, features are switched off in order, from the least to the most important, each with a switch (a **kill switch**) that works in seconds. The two numbers watched live are the ones viewers feel: **start time** (seconds from pressing play to picture) and **rebuffering ratio** (share of watching time spent frozen).

1	Stop 4K and 1080p on phones	saves the most bandwidth; few notice on a small screen
2	Pause animations, live polls, comments	heavy on servers, light on value tonight
3	Show a fixed home page, not a personal one	no recommendation calls for new arrivals
4	Queue new sign-ups in a short waiting room	protects viewers already watching
5	Cap every stream at 480p	last resort: blurrier for all, frozen for none
X	Never: stop the live match	the one thing everyone came for

Load rising ↓ · Each step has a switch that works in seconds and was tested in rehearsal.

Figure 6.46 · The ladder for final night. Each step saves load and costs little that viewers notice. The live match is the last thing that may ever be touched. *Author's illustrative ladder; order agreed before the match, switches tested in rehearsal.*

PM decision: the order of the ladder is a product decision: which feature matters least to viewers and to revenue tonight? The PM writes it down before the match, because at 9:45 pm there is no time to debate it.

8. On-demand is different

Everyday picture. Bread for tomorrow can be baked and delivered to shops tonight. Only fresh milk has to move at dawn.

How it works. For films and series, nobody needs the same second at the same time, so platforms **place copies in advance**: overnight, when networks are quiet, caches are filled with what people will probably watch tomorrow (a new season, today's Top 10). Viewing is lopsided: a small share of titles gets most views, so caching the most popular few per cent covers most requests. **Downloads** let viewers save episodes on home Wi-Fi and watch on the train without mobile data. **PM decision:** which titles to place where, and whether to push downloads for markets with costly data, such as mobile-first India.

9. DRM and piracy

Everyday picture. A sealed milk pack with a batch number: if adulterated milk turns up, the batch shows where it came from.

How it works. Video segments are encrypted; a player gets the key from a **licence server** only after the app checks the viewer's plan and device. This is **DRM** (digital rights management), built into Android, iOS and TVs (Google's Widevine, Apple's FairPlay). DRM can't stop someone filming a screen, so platforms add an invisible **watermark**, a tiny pattern unique to each account, woven into the picture. When a pirated IPL stream appears on Telegram, the watermark shows which account it came from, and that account can be cut off within minutes. **PM decision:** stronger protection (highest quality only on the most secure devices) versus reach (older phones and cheap TVs). Rights holders often decide it in the contract: 4K only with the strongest DRM.

Final night: the plan on one page

When	What happens	Why
The week before	Forecast the peak from past finals; book CDN capacity with two providers; rehearse the degradation ladder with real switches	Capacity can't be bought in the last minute
Match day, noon	Freeze app releases; pre-warm servers for the forecast plus a margin; warm caches with the home page and ads	No surprises from new code; ready before the crowd
7:00–7:35 pm	Let logged-in phones skip login; let viewers watch while payments complete; spread ad requests over time	The first wave hits login and payments, not video
During play	Watch start time and rebuffering by city and network; move traffic between CDNs if one struggles	These are what viewers feel
If load passes the limit	Walk down the ladder: 4K, then extras, then recommendations; never the match	Save the one thing everyone came for
After the final	Scale down in stages; review every alarm; show the "after the IPL" watchlist	The next morning is when renewals are won or lost

IN THE INTERVIEW · HOW STREAMING SYSTEMS COME UP

- **"How would you handle 6 crore concurrent viewers?"** Bitrate ladder and ABR, CDNs with a high hit rate, pre-warming before the toss, and a degradation ladder that never touches the match.
- **"Why is the app 30 seconds behind TV?"** Segments and the safety buffer; shrinking them lowers delay but raises freezes on weak networks.
- **"Which metrics would you watch during the final?"** Start time and rebuffering ratio by city and network, plus login and payment success in the first half-hour.

Connect the dots

Four ideas ran through the cases, the products and the technology. A PM at a streaming company meets all four in the same week, often on the same final night.

Thread	In the cases	In the products	In the technology
Pay once, earn many times	Keeping an IPL joiner costs no extra content; ₹105 Cr a quarter from a 10-point lift in renewals	Netflix's flywheel; ₹58 Cr of rights per IPL match; a game player must be worth more than the \$3 it cost to win them	A CDN copy fetched once serves a whole city; a 99% hit rate means five times less load than 95%
Events rent attention; a library keeps it	The June cliff in the IPL cohort; light viewers drifting to free films	JioHotstar's serials after cricket; Prime Video keeping Prime renewing; Spotify's Discover Weekly habit	Live needs pre-warming and a degradation ladder; on-demand is placed in caches the night before
Count who moves down	A ₹99 slot wins more sharers but pulls ₹499 homes to ₹298; views fell because people, not minutes, went missing	Family plans pay only where families had one payer; creators follow the audience, not the highest share	The player steps down a rung before the buffer empties: blur is better than a freeze
Trust the record	Check what exists today before recommending it (X-Ray, rentals, no extra member in India)	Content ID protects rights owners; royalties are split by share of streams	A blockchain makes a record tamper-evident; DRM and watermarks trace a leak to one account

BACK TO AHMEDABAD · A SKETCH ANSWER FOR THE FINAL-NIGHT QUESTION

The eight seconds. A slow start on the busiest night of the year is a solved kind of problem. It has a playbook: servers switched on before the toss, logged-in phones skipping login, a player that starts on a low rung and steps up, and a ladder of features to turn off before the match is touched. Eight seconds to start, then a blurry picture that sharpens, is the system working as designed. A freeze in the middle of the last over would be far worse, and that is what start time and rebuffering ratio are watched for, city by city.

The June cancellations. This is the bigger business problem. In the illustrative cohort, about two in three IPL joiners leave within a month of the final. Lifting that renewal by ten points is worth about ₹105 crore a quarter, with no new content cost. The fixes are product work: a pause instead of a cancel, an “after the IPL” watchlist built during the season, and a calendar of the next cricket.

So which worries more? The June cancellations, because they repeat every year and cost money quietly. The buffering matters as a condition: nobody renews an app that failed them during the final. **Goal (North Star):** IPL joiners still watching weekly 60 days after the final. **Safety checks (guardrails):** start time and rebuffering ratio on final night; revenue per subscriber. **Trade-off:** promoting serials during matches earns tomorrow's renewals but takes screen space from tonight's ads.

KEY IDEA · PAY ONCE, THEN EARN BETWEEN THE BIG MOMENTS

Streaming and music businesses pay a large, fixed sum for what people want, then need as many people as possible, for as long as possible, to pay for it through plans, ads or bundles. Big moments (a final, a new season) bring people in. What happens between the moments decides the business. When you design, price or measure anything here, ask: what brings people in, what keeps them once it ends, and who will move to a cheaper option if we offer one?